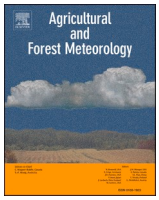
Agricultural and Forest Meteorology 382 (2026) 111131



Contents lists available at ScienceDirect

Agricultural and Forest Meteorology

journal homepage: www.elsevier.com/locate/agrformet





Developing and testing of a winegrape phenology model for Australian conditions

Dane S. Thomas [a,c], Wendy Cameron [b,*], Peter T. Hayman [a,c,d], Marcos Bonada [a,d], Victor O. Sadras [a,c,d], Joanna M. Gambetta [e], Jessica Y.C. Jolley [f,g], Paul R. Petrie [a,c,d,h]

[a] Crop Sciences, South Australian Research and Development Institute, Waite Campus, Adelaide, Australia
[b] School of Agriculture, Food and Ecosystem Sciences, University of Melbourne, Parkville, Australia
[c] School of Agriculture, Food and Wine, University of Adelaide, Waite Campus, Adelaide, Australia
[d] College of Science and Engineering, Flinders University, Adelaide, Australia
[e] School of Environmental and Life Sciences, College of Engineering, Science and Environment, University of Newcastle, Newcastle, Australia
[f] Food Sciences, South Australian Research and Development Institute, Waite Campus, Adelaide, Australia
[g] School of Animal and Veterinary Sciences, University of Adelaide, Roseworthy Campus, Adelaide, Australia
[h] School of Mechanical and Manufacturing Engineering, University of New South Wales, Sydney, Australia

ARTICLE INFO

Keywords:
Vitis vinifera
Phenology
Model

ABSTRACT

Within season vine management and longer-term strategic decisions for vineyards and wineries require an understanding of vine phenology and tools for its prediction. Using a curated dataset of 3,159 phenological observations for ten varieties and across seventeen regions, the largest reported Australian phenological data base, we aimed to develop phenological models for budburst, flowering, veraison and maturity for an Australian context. Published winegrape phenology models were initially tested with the Australian dataset with poor results. These models were then improved by examining different methods to calculate daily forcing units (F*) and alternative start day of accumulation of forcing units and then modelling a phenophase by commencing the accumulation of forcing units from an earlier phenophase. A range of single temperature function models were developed and evaluated. These models improved predictions, particularly when in season phenology observations were available to update the predictions through the season. The models we developed are robust, relatively easy to use, and better suited to an Australian context. Phenological models provide a useful tool to assist vineyard planning and therefore efficiency in the short term and contribute to long term planning in the vineyard and winery, particularly in a variable and changing climate.

1. Introduction

Within season vine management and longer-term strategic decisions for vineyards and wineries require a good understanding of crop phenological development and predictive tools, to optimise the timing of pesticide or fertiliser application, irrigation, or logistics for harvest (Fraga et al. 2016, Wang et al. 2020, Essling et al. 2021). Vineyard and winery planning is complicated by a variable and changing climate and the resulting shifts in the timing of all phenophases. Not only is maturity occurring earlier (Petrie and Sadras 2008, Webb et al. 2011, Cameron et al. 2020), there are increased infrastructure requirements due to compression of the harvest period (Petrie and Sadras 2016). Other phenological phases, flowering and veraison, are also advancing (Duchêne and Schneider 2005, Webb et al. 2007, Petrie and Sadras 2008, Duchêne et al. 2010, Tomasi et al. 2011, Ruml et al. 2016). We note that some observed advancement trends for maturity, veraison and flowering are not significant, and some non-significant trends for a delay in these phenophases have been reported such as that for maturity in the Margaret River, Australia (Webb et al. 2011). Trends for budburst are less defined with advancement (Duchêne and Schneider 2005, Bock et al. 2011, Bernáth et al. 2021, Cameron et al. 2022), delay (Malheiro et al. 2013) or no change (Tomasi et al. 2011) being reported.

Phenological models facilitate day to day, on the ground, vineyard management decisions. For example, being able to predict harvest time is crucial, not only to ensure that any required pesticides are not applied within withholding periods, but also to allow organisation of staff and

* Corresponding author.
E-mail address: wendy.cameron1@unimelb.edu.au (W. Cameron).

https://doi.org/10.1016/j.agrformet.2026.111131
Received 2 September 2025; Received in revised form 9 March 2026; Accepted 14 March 2026
Available online 18 March 2026
0168-1923/© 2026 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

other winery and vineyard equipment and logistics. Knowledge of the timing of major phenophases is also vital when planting new vineyards, to help determine which varieties are best suited to the climate (Van Leeuwen et al. 2024).

Phenological models are important to understand the response and implications of climate change scenarios (Chuine 2000, Basler 2016, Parker et al. 2020a). The suitable model will depend on the phenological phase being modelled and the plant species, but temperature is the main variable (Chuine et al. 1999). The earliest phenological models were simple temperature accumulation models (Wang 1960, Mo et al. 2023), and many similar "thermal time" models have been developed since, which are mostly linear and single phase (Mo et al. 2023). Other phenological models, such as those using a non-linear, sigmoid approach to temperature accumulation (Caffarra and Eccel 2010), are more complicated, and take into account that plant growth rate may, for example, slow at increased temperatures. Models may also use a two-phase approach, such as those for budburst, where a chilling accumulation is satisfied in conjunction with a warming (forcing) temperature accumulation. These two-phase models may be sequential i.e., chilling assessed before warming (Hunter and Lechowicz 1992, García De Cortázar-Atauri et al. 2009), parallel i.e., chilling and warming considered together (Hänninen 1990, Kramer 1994) or alternating, which assumes warming (forcing) requirement is decreased when chilling increases (Murray et al. 1989, Kramer 1994, Mo et al. 2023). More complex phenological models have introduced photoperiod (Caffarra et al. 2011). In a changing climate, the asymmetry of day and nighttime temperatures, possible latitude effects, with high-latitude areas warming faster than low latitude areas (Parmesan 2007, Mo et al. 2023), and the impact of other climatic variables can influence models. Despite the potential complexity of phenological models, ultimately, model accuracy, parsimonious data requirement and ease of use are required for their practical application.

For grapevines, a number of phenological models have been published. Typically, these models predict development using a temperature function to describe either chilling or heat forcing units (F*), with the model detailing the number of units required for a particular phenological phase to be reached. Well established grapevine models such as van Leeuwen et al. (2008), García De Cortázar-Atauri et al. (2009) and Parker et al. (2011), are driven by empirical responses to temperature and exclude factors such as water stress, crop load or pruning. One potential problem with temperature models however, is that they may not adequately reflect temperature thresholds, particularly on the higher end of the spectrum (García De Cortázar-Atauri et al. 2017) that may limit grapevine growth (Kriedemann 1968). These higher temperatures may need more consideration in the development of phenology predictive models, particularly in a warming climate. Another concern is that the existing published models have been developed using phenological data from smaller geographical areas or areas remote from where the models might be applied. Most of these models have been developed using European data. For example, the García De Cortázar-Atauri et al. (2009) models for budburst were developed using phenological data for locations in France. Parker et al. (2011) developed models using a broader set of data from France, Switzerland, Italy and Greece, but data were predominantly from France. It is unclear whether these models can be used to predict phenology in other grapegrowing regions or countries.

There are approximately 6,000 grapegrowers and 2,100 wineries in Australia. The industry employs around 200,000 people and contributes about $51.3 billion annually to the Australian economy (Wine Australia 2024). The 65 regions cover diverse climates from hot and semi-arid, warm summer mediterranean, through to oceanic (Iland et al. 2017). Our study aims to determine firstly whether published models are applicable to phenological data observed in Australia, and secondly, if required, to improve these models to suit the Australian environment. The model development phase was to update the critical sum of the daily forcing units (F*) required to reach a phenophase for the already published models. Further improvements examined different methods to calculate daily forcing units and alternative start days from when the accumulation of forcing units commenced and finally, modelling a phenophase by commencing the accumulation of forcing units from an earlier, actual phenophase day.

2. Materials and methods

2.1. Phenological data

As there is no national repository of Australian phenological data, the data for this study were obtained from published research trials (393 observations) (McCarthy 1997, Sommer et al. 2000, Goodwin 2009, Holzapfel 2009, Tyerman 2015, Treeby 2018, McCarthy 2019, Walker et al. 2020), observational records from research trials (597 observations) (Sadras et al. 2009, Sadras and Moran 2013, Petrie et al. 2017, Moran et al. 2019, Bonada et al. 2020, 2022), and from commercial vineyards (2169 observations) (data provided in confidence). Ten varieties and seventeen wine regions were represented (Table 1, Supplementary Table 1S), and as such, this is the largest reported Australian phenology data set. These wine regions cover a large geographical area in Southeastern Australia and a range of climates from cool (e.g., Grampians) to hot (e.g., Riverland) (Wine Australia 2025).

Our phenological observations spanned vintages from 1983 to 2022; 470 observations were from vintages from 1983 to 2003 and 2689 were from vintages from 2004 to 2022. Most observations were from Shiraz (872), Chardonnay (590), Cabernet Sauvignon (457) and Merlot (334), with a moderate number for Pinot Noir (249) and Sauvignon Blanc (221), and smaller numbers of observations for Pinot Gris (117), Semillon (112), Riesling (106) and Cabernet Franc (101) across the four phenological phases (Table 1). Chardonnay, Shiraz, and to a lesser extent Cabernet Sauvignon and Merlot, had observations from many Geographic Indications (GI's) whereas the other varieties had observations from between 2 and 4 GI's (Supplementary Table 1S).

Phenological assessments were made using EL phases in the Eichorn and Lorenz (1977) scale. A phenological phase was taken to have been achieved when it had been reached by 50% of the organs being assessed, i.e. budburst when 50% of buds had reached EL4, flowering when 50% of flowers were open (EL23), veraison when 50% of berries were soft or coloured (EL35). The dates for these phenological phases were taken directly from the published reports and commercial vineyard data, or if necessary, were calculated from the given EL phase by linear interpolation of their weekly or fortnightly inspections. The dates of the phenological phase used in this study are subject to a degree of uncertainty associated both with potential errors in the estimation of a phase from visual observations of the individual vines or section of the vineyard examined, and by the uncertainty that these vines are a true representation of the vineyard or region. A further source of uncertainty is the assumption that the numerical EL phase is linear with time and that therefore the targeted EL phase (e.g. EL23 or EL 35) can be interpolated from observations collected at approximately 1-2 week intervals (Sadras and Moran 2013).

The pseudo-phase of harvest (Menzel et al. 2006) was taken to be the date when the fruit was picked. The date of harvest is often a commercial and/or operational decision and does not necessarily correspond to biological or oenological traits such as sugar concentration or other quality characteristics. The scarcity of sugar observations in the data set precluded using this trait to define maturity (Parker et al. 2020a).

2.2. Weather data

Daily minimum and maximum temperatures used in the modelling were obtained for the 0.05° grid (approximately 5km grid) surrounding each vineyard from the SILO database as patched point data (Jeffrey et al. 2001). These daily temperatures are thus an approximate estimate for the actual location of the vineyard. Mean temperatures were

Table 1
Number of observations of the main phenological phases for ten *Vitis vinifera* varieties and number of observations for instances where two main phenological phases occurred (used for within season updating models).

Variety	BB (EL4)	FLO (EL23)	VER (EL35)	HAR (EL38)	BB and FLO	BB and VER	BB and HAR	FLO and VER	FLO and HAR	VER and HAR
Shiraz	208	230	226	208	188	184	165	207	172	168
Chardonnay	117	152	165	156	96	107	97	142	126	134
Cabernet Sauvignon	123	122	112	100	99	98	79	95	77	71
Merlot	80	103	85	66	77	65	46	82	59	46
Pinot Noir	43	71	77	58	42	41	36	69	55	56
Sauvignon Blanc	42	64	66	49	39	42	32	60	45	48
Pinot Gris	17	35	38	27	17	17	12	35	26	27
Semillon	29	35	32	16	28	27	11	31	15	12
Riesling	27	35	29	15	26	24	8	29	15	9
Cabernet Franc	29	30	31	11	28	28	10	30	11	11
Total	**715**	**877**	**861**	**706**	**640**	**633**	**496**	**780**	**601**	**582**

Note: Table 1 is presented in order from the variety with the most observations overall (Shiraz) to the variety with the least observations (Cabernet Franc). EL; Eichorn Lorenz phases of phenology (Eichorn and Lorenz 1977). BB; budburst, FLO; flowering, VER; veraison, HAR; harvest.

calculated from daily minimum and maximum temperature.

2.3. Model development

The steps taken to develop our phenological model are summarised in Table 2. Initially we evaluated existing published models for each variety using their F* (Table 2, step 2.3.1), then re-assessed these models using F* that were updated using the Australian data (2.3.2). These published models had mostly linear structures using growing degree days (GDD), with some different start dates and/or base temperatures. We then used a range of temperature accumulation methods (linear, curvilinear, sigmoid) to parameterise and validate across-variety (species) level models for all phenophases (2.3.3). The optimal temperatures and start days obtained from these models were then simplified, for example, rounded to the nearest 5°C or to a day representing the beginning of a month (2.3.4), and the models were re-evaluated. The best of each temperature accumulation model type was selected to then parameterise and validate variety level models (2.3.5). Finally, we evaluated these models using actual phenophase dates to predict a subsequent phenophase date within the same season (2.3.6).

2.3.1. Evaluate published phenological models

The complete observational phenological dataset (Table 1) was used to test selected published models (Table 3). The basis of selection of the models was that they were published relatively recently and were developed using a range of varieties over a number of sites. The Australian model Vinelogic (Godwin et al. 2002) was not included because we were not permitted to publish the mathematics of that model. The models selected on this basis, are similar in many aspects. The van Leeuwen et al. (2008) model is a single phase model that uses a

Table 2
Summary of steps taken during phenological model development.

Step	Description of process	Phenophase modelled	Level of model	Mathematical function used to calculate heat units	Start day, Temperature thresholds	F* used	Data	Statistical Metrics	Software
2.3.1	Evaluate published phenological models	Model dependent	Variety	Fixed (by literature)	Fixed (by literature)	Fixed (by literature)	Complete observational dataset for all varieties (Table 1).	RMSE, MBE, EF	R
2.3.2	Update F* of published models, then re-evaluate	Model dependent	Variety	Fixed (by literature)	Fixed (by literature)	Fitted	Data split for each variety. Half for parameterisation, half for validation.	Parameterisation: RMSE, EF Validation: RMSE, EF, MBE	R-LOOCV
2.3.3	Develop across-variety (species) level models.	BB, FLO, VER, HAR	Species	GDD, Richardson, Sigmoid, Wang and Engel, BRIN	Fitted	Fitted	Data split, half parameterisation, half validation using observations for all varieties.	Parameterisation: AIC, RMSE, EF Validation: RMSE, EF, MBE	PMP
2.3.4	Develop simplified across-variety (species) level models.	BB, FLO, VER, HAR	Species	GDD, Richardson, Sigmoid, Wang and Engel, Single Sine	Fixed (by us to "neat" values day 1, day 32, 5°C, 10°C etc informed by fitted values)	Fitted	Data split, half parameterisation, half validation using observations for all varieties.	Parameterisation: AIC, RMSE, EF Validation: RMSE, EF, MBE	R-LOOCV
2.3.5	Develop variety level models (a selection of the better models from above – based on AIC)	BB, FLO, VER, HAR	Variety	As in 2.3.4	Fixed (by us to "neat" values day 1, day 32, 5°C, 10°C etc informed by fitted values).	Fitted	Data split for each variety. Half for parameterisation, half for validation.	Parameterisation: RMSE, EF Validation: RMSE, EF, MBE	R-LOOCV
2.3.6	Develop within season updating models using actual previous phenophase timing as start day	FLO, VER, HAR	Variety	As in 2.3.4	Fixed (by date of previous phenophase or 5°C, 10°C etc)	Fitted	A subset of the complete dataset used. This data was split for each variety. Half for parameterisation, half for validation.	Parameterisation: RMSE, EF Validation: RMSE, EF, MBE	R-LOOCV

BB: Budburst, FLO: Flowering, VER: Veraison, HAR: Harvest. AIC: Akaike information criterion, RMSE: root means squared error, MBE: mean bias error, EF: model efficiency. See section 2.4 for formulas for temperature functions and 2.5 for Statistical methods.

Table 3
Published models that were evaluated using a large data set of Australian observations. Start day for models developed in the Northern Hemisphere were converted to Southern Hemisphere dates. The model codes shown here are used extensively in the Figures and text. They indicate the model type, base temperature (if applicable), start day and model author abbreviation respectively. GDD: growing degree days, Tbase: base temperature.

Author	Model code	Phase modelled	Model	Start Date, Days from 1st July
Van Leeuwen et al. (2008)	GDD 10, 1 VL	Budburst	GDD. Tbase = 10°C	1st July, 1
García de Cortázar-Atauri et al. (2009)	GDD 10, 1 GCA	Budburst	GDD. Tbase = 10°C	1st July, 1
García de Cortázar-Atauri et al. (2009)	GDD 5, 1 GCA	Budburst	GDD. Tbase = 5°C	1st July, 1
García de Cortázar-Atauri et al. (2009)	BRIN GCA	Budburst	BRIN chill then heat	1st February, -150 or -151
Van Leeuwen et al. (2008)	GDD 10, 1 VL	Flowering	GDD. Tbase = 10°C	1st July, 1
Parker et al., (2013)	GDD 0, 60 PK	Flowering	GDD. Tbase = 0°C	29th August, 60
Van Leeuwen et al. (2008)	GDD 10, 1 VL	Veraison	GDD. Tbase = 10°C	1st July, 1
Parker et al., (2013)	GDD 0, 60 PK	Veraison	GDD. Tbase = 0°C	29th August, 60
Parker et al., (2020a)	GDD 0, 91 PK	Maturity	GDD. Tbase = 0°C	30th September,91
Parker et al., (2020a)	SIG 86 PK	Maturity	Sigmoid	25th September, 86
Van Leeuwen et al. (2008)	GDD 10, 1 VL	Harvest	GDD. Tbase = 10°C	1st July, 1

linear growing-degree day (GDD) temperature accumulation function with a base temperature (Tbase, Tb) of 10°C and start day 1st July (all dates are for the southern hemisphere), i.e. day 1, for all phenology phases (GDD 10, 1 VL). The Parker et al. (2013) Grapevine Flowering Veraison (GFV) model, evaluated here for flowering and veraison, uses the same linear, single phase GDD temperature accumulation function, but with a base temperature of 0°C and a different start day of 29th August, i.e. day 60 (GDD 0, 60 PK). The García De Cortázar-Atauri et al. (2009) models evaluated budburst only. Similarly to van Leeuwen et al. (2008) and Parker et al. (2013), two use a GDD function with base temperature of either 10°C or 5°C and start day of 1st July (GDD 10, 1 GCA; GDD 5, 1 GCA). The third option for the García De Cortázar-Atauri et al. (2009) model used for budburst was the BRIN (BRIN GCA) model. It is a two-phase, sequential model that uses Bidabe's Cold Action dormancy model (García De Cortázar-Atauri et al. 2017) to calculate the dormancy period, followed by Richardson et al. (1974, 1975) model to calculate the post-dormancy period. Finally, the maturity models of Parker et al. (2020a) were used to examine day of harvest. The first of these includes the same GDD accumulation, but with a base temperature of 0°C and a start day of 30th September, i.e. day 91 (GDD 0, 91 PK). The second Parker et al. (2020a) maturity model uses a nonlinear, single phase sigmoid temperature accumulation function commencing on 25th September i.e. day 86 (SIG 86 PK) that limits the effectiveness of higher temperatures. Parker et al. (2020a) investigated the required critical forcing units for a range of harvest sugar concentrations and showed that critical forcing units increased as berry sugar concentration increased. We used the critical forcing unit (F*) corresponding to the highest sugar concentration provided by Parker et al. (2020a) to predict the day of harvest.

The selected models were evaluated using linear regression in the R framework (R Core team, 2018). The F* critical (F*) used for this initial evaluation of the models were those from the published models. The statistical metrics examined are detailed in Table 2 and section 2.5 Statistical Methods.

2.3.2. Update F* of published models

We then sought to improve these published models by updating the critical number of forcing units (F*) for these models at a variety level. We did not update temperature thresholds in this instance as our assumption was that these published models were robust and we wanted to understand whether the models were transferable with just a localised F* and the original temperature thresholds. In later steps we updated temperature thresholds. For this step, the dataset for each variety was split in two halves, one for parameterisation and one for validation. Each of these half-datasets contained 50 or more observations of each main phenological phase for Shiraz, Chardonnay and Cabernet Sauvignon; between 20 and 50 for Merlot, Pinot Noir and Sauvignon Blanc and fewer than 20 observations of each main phase for other varieties. As there were fewer observations available for Pinot Noir, Sauvignon Blanc, Pinot Gris, Riesling, Semillon and Cabernet Franc (Table 1), we did not feel these results were as robust. Results for those varieties for all steps of model development have been provided in Supplementary data for completion but are not discussed.

The parameterisation dataset was used to update F* using leave-one-out-cross-validation (LOOCV) in the Caret package within the R framework (Kuhn 2008). These models were then validated using the validation dataset. Statistical criteria for parameterisation and validation are indicated in Table 2.

2.3.3. Develop across-variety (species) level models

As a first step in developing a model suited to Australian conditions, in similarity with Parker et al. (Parker et al. 2011, 2013, 2020a), across-variety (species) level models were initially evaluated across all varieties, prior to parameterising the critical number of forcing units (F*) for each variety. These models were initially parameterised using the Phenological Modelling Platform (PMP) software, version 5.5 (Chuine et al. 2013) using the simulated annealing algorithm of Metropolis (Metropolis et al., 1953) to optimise the parameters of different temperature functions. Optimisation parameters were those recommended by the PMP software, and parameter ranges for fitted variables were not set. A minimum of 10 repetitions of the parameterisation were performed with the final model selected on the basis of lowest RMSE when fitted parameters made physiological sense. Initially all parameters (threshold temperatures and start day) were fitted for five temperature functions; GDD, Richardson, Sigmoid and Wang and Engel; the method used to calculate heat forcing units for these various temperature functions is described in section 2.4. These temperature functions differ in threshold temperatures and their mathematical approach to accumulating temperature (linear, sigmoid or curvilinear) (Section 2.4). The two-phase BRIN model was also included at this step for comparison. These across-variety (species) level models were evaluated using the validation datasets, which were the union of validation datasets for all ten varieties. Evaluation metrics are indicated in Table 2.

2.3.4. Develop simplified across-variety (species) level models

To determine if a more generalised model that used the same heat function, but with a different start day or threshold temperature for different phenological phases was sufficiently robust, models were developed when (i) start day was fixed at either day 1 (1st July), day 32 (1st August) or day 63 (1st September) (dates chosen for simplicity); and (ii) when threshold temperatures were fixed to rounded values such as 0, 5 or 10°C in similarity to García De Cortázar-Atauri et al. (2009) and Parker et al. (2020a). The PMP was necessary to establish temperature thresholds, start days and F* in the previous step. We reverted to the R platform once we had obtained the optimal temperatures and start days

from the PMP. This enabled us to include a fifth temperature function, the Single Sine (see Section 2.4), which uses both maximum and minimum temperatures, but which was unable to be included in the PMP step (2.3.3) due to its requirement for two temperature parameters. We could include Single Sine for this step of the evaluation using the R framework (R-Core-Team 2018). A temperature function such as Single Sine is of particular interest with respect to the effect of climate change on phenology in light of emerging evidence of the importance of day and nighttime temperatures and asymmetric warming (Mo et al. 2023). The same evaluation process for these models was used as in Section 2.3.2 (Table 2).

2.3.5. Develop variety level models

The model frameworks (function, threshold temperature(s), start day) of better performing simplified across-variety (species) level models (see above – Section 2.3.4) were then used to parameterise the critical number of forcing units (F*) at a variety level for Cabernet Sauvignon, Chardonnay, Merlot and Shiraz. These models were evaluated as above in Section 2.3.4 (Table 2).

2.3.6. Develop within season updating models

Instead of using a set start day, we considered the use of actual dates of a phenological phase as they occurred within the season, to model a subsequent phenological date within that same season. This represents a potential use of phenological models in an operational context. We had a subset of the observation dataset with dates for two phenophases within the same season (Table 1). These data were used to develop phenology models that could serve as updating models within a season by commencing the heat accumulation from the observed date of the earlier phenology phase. This restricted dataset for each phenology period was randomly divided, half for parameterisation and half for validation for each variety. The parameterisation of F* and the validation was undertaken in the same way as 2.3.5 above but with the restricted dataset. Statistical evaluation metrics are shown in Table 2.

2.4. Temperature response functions for phenology models

Temperature response functions commonly describe the accumulation of heat forcing units (or degree days) based on mean daily temperature. The functions commence accumulating this daily production of heat forcing units starting from a given day and the phenological phase is predicted to occur when the sum of the daily forcing units reaches a critical value (F*).

Several temperature functions were used to calculate heat forcing units. Often there are limits placed on these temperature functions, such as the use of base temperature limiting cooler temperatures, acknowledging that growth does not occur below this base temperature, or an upper or maximum temperature limiting warm temperatures, recognising that plant growth is restricted at high temperatures. Others are non-linear functions where the effectiveness of temperature on growth varies as the temperature increases. As such, the different temperature response functions will provide different temperature summations for the same data set.

The different temperature response functions, i.e. methods of heat summation, that were examined were:

i) Growing degree days (GDD)

This function has been widely used to describe the phenology of species including grapevines (García De Cortázar-Atauri et al. 2009, Parker et al. 2011, 2013, 2020a, Wang et al. 2020). The GDD function is linear and calculates the cumulative daily mean temperature (Td) above a temperature threshold (Tbase (Tb), base temperature) usually starting from a given date. The phenological phase occurs when a critical sum of forcing units is reached. It assumes that once the temperature is greater than the base temperature, that effect of temperature on growth is linear, regardless of temperature, with no upper limit.

$$fGDD = Max(0,\ Td - Tbase) \tag{1}$$

i) Richardson

This function is a modified version of the GDD function that includes a base temperature (Tbase) and a plateau above the threshold parameter, Thigh (Richardson et al. 1974, 1975). Richardson is therefore linear and similar to GDD except there is an upper temperature limit placed on the temperature accumulation.

$$fRichardson(Td) = Max(Min(Td - Tbase,\ Thigh - Tbase), 0) \tag{2}$$

i) Sigmoid

The sigmoid function is non-linear and has two parameters, a, which defines the sharpness of the response function (values away from zero induce a sharper response curve) and b, the mid-response temperature. The shape of the sigmoid curve determines at which temperatures the rate of heat summation slows or is optimum.

$$f\ \text{Sigmoid}\ (\text{T}d) = \frac{1}{1\ + \exp[\text{a}(\text{Td}\ - \text{b})]} \tag{3}$$

i) Wang and Engel

This non-linear function was first defined by Wang and Engel (1998). It has an optimum temperature and is not symmetrical in its heat accumulation rate on either side of the optimum temperature. It has three parameters: a base temperature (Tbase), an optimum temperature (Topt) and an upper threshold (Thigh) above which the heat accumulation equals zero. The formula below is correct, but in the pdf proof part of the formula has been cut off i.e. it does not fit in the column and has just been cut off. In the other words, the formula in the pdf proof is incomplete. The formula below (4) is correct and we need to ensure it is complete in the pdf version of the article.

$$f\ Wang\&Engel(Td) = Max\left(\left(2(Td - Tbase)^{\alpha}(Topt - Tbase)^{\alpha} - \left(\frac{(Td - Tbase)^{2\alpha}}{(Topt - Tbase)^{2\alpha}}\right), 0\right)\right. \tag{4}$$

where:

$$\alpha = \frac{\ln(2)}{\ln(Thigh - Tbase)} \Big/ (Topt - Tbase))$$

i) Single Sine

This technique uses daily minimum and maximum temperatures, rather than mean temperature to produce a sine curve over a 24-hour period. It assumes the temperature curve is symmetrical around the maximum temperature and then estimates degree days for that day by

calculating the area beneath the temperature curve, between the lower (Tlower) and upper thresholds (Tupper). The equations can be found in (Zalom et al. 1983). For example, this technique has been used by Tombesi et al. (2010) and Ortega-Farias and Riveros-Burgos (2019).

2.5. Statistical methods

The performance of models were assessed on the criteria: (i) Akaike Information Criterion (AIC) (Burnham and Anderson 2002) which rates models in terms of parsimony and efficiency, where the lowest value is associated with the best model; (ii) the root means squared error (RMSE), which gives the mean error of the prediction in days; (iii) the model efficiency (EF) which explains the percentage of variance explained, where a negative value indicated that the model performed worse than the null model (mean date of the phenology observations (Parker et al. 2011, Basler 2016)), and a value above zero indicates that the model explained more variance than the null model (with a maximum value of 1); and (iv) the mean bias error (MBE), which gives the average predicted error representing the systematic error of the model. The smaller the value, the smaller the bias. Positive MBE indicate the predicted values are, on average, larger than actual values (i.e., later dates) and negative MBE values indicating predicted values are, on average, smaller than actual values (i.e., earlier dates). Equations for these criteria are shown below:

$$AIC = n \times ln\left(\frac{\sum_{i=1}^{n}(S_i - O_i)^2}{n}\right) + 2k \quad (5)$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(S_i - O_i)^2}{n}} \quad (6)$$

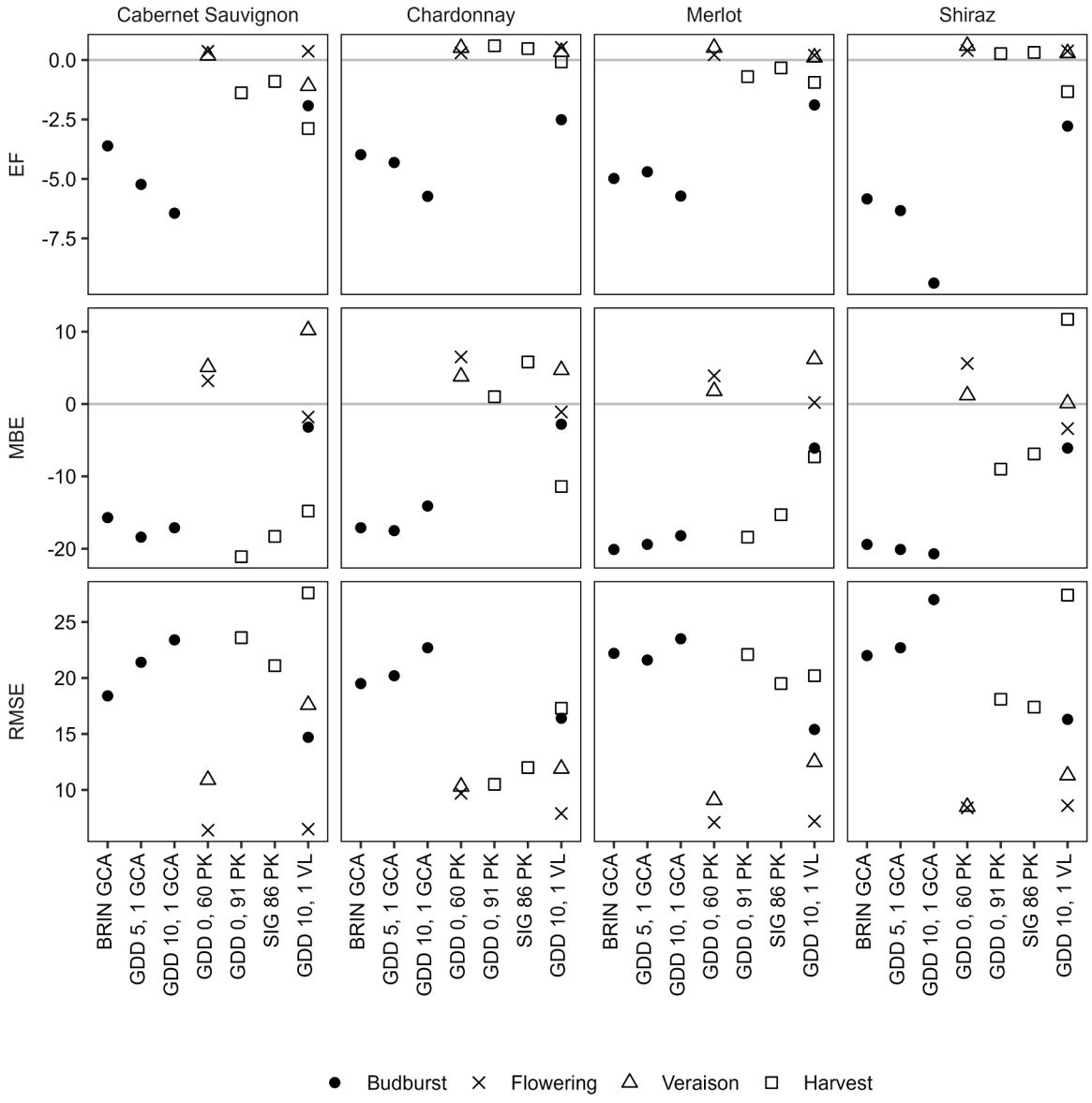


Fig. 1. Performance of published models predicting major phenophase timing using the published F* with an Australian dataset and four varieties, Cabernet Sauvignon, Chardonnay, Merlot and Shiraz. The models were those described by van Leeuwen et al (2008) GDD function Tb = 10°C start day 1 (GDD 10,1 VL) for all phenophases; Garcia de Cortazar-Atauri et al (2009) GDD function Tb = 10°C, commencing at day 1 (GDD 10, 1 GCA); Garcia de Cortazar-Atauri et al (2009) GDD function Tb = 5°C, commencing at day 1 (GDD 5, 1 GCA) and Garcia de Cortazar-Atauri et al (2009) BRIN function commencing on day -150 where critical chill (Cc) is satisfied before critical heat (Gc) commences (BRIN GCA) for budburst. Parker et al (2013) GDD Tbase 0°C commencing day 60 model (GDD 0, 60 PK) for flowering and veraison. Parker et al (2020a) GDD Tbase 0°C start day 91 (GDD 0, 91 PK), and Parker et al (2020a) Sigmoid model start day 86 (SIG 86 PK) for harvest. EF; model efficiency, RMSE; root mean square error (days), MBE; mean bias error (days).

$$EF = 1 - \left(\frac{\sum_{i=1}^{n} (S_i - O_i)^2}{\sum_{i=1}^{n} (O_i - \overline{O})^2} \right) \quad (7)$$

$$MBE = \frac{\sum_{i=1}^{n} (S_i - O_i)}{n} \quad (8)$$

Where S denotes predicted and O denotes observed values, $\overline{O}$ is mean of observations, and n is the number of observations, and where k is the number of parameters required to fit the model

3. Results

3.1. Evaluate published phenological models

The performance of several published phenology models (Table 3) was assessed with the Australian dataset, using their published F* values (Supplementary Table 2S).

3.1.1. Budburst for published phenological models

The published models predicted budburst poorly with large RMSE, poor efficiency and a large negative bias of up to -20.7 days for Australian observations (Figure 1). The importance of F* can be seen where the García de Cortázar-Atauri et al. (2009) and van Leeuwen et al. (2008) models that both used GDD Tbase = 10°C and start day 1st July (that is, GDD 10, 1 VL and GDD 10, 1 GCA) gave quite different results (Figure 1).

The EF values were negative for all models tested for budburst, for all varieties (Figure 1), indicating a worse performance than the null model i.e., predictions based on the mean date of budburst (Parker et al. 2011, 2013). EF were smallest (least negative) for the GDD 10, 1 VL model, and largest for the equivalent GDD 10, 1 GCA model, attributed to the difference in their F* values (Table 2S).

Reflecting the bias and EF, RMSE were lowest for the budburst model GDD 10, 1 VL (range 14.7 – 16.4 days), and highest for the equivalent GDD 10, 1 GCA model (range 22.7 – 27 days) for the four varieties; Cabernet Sauvignon, Chardonnay, Merlot and Shiraz (Figure 1).

3.1.2. Flowering for published phenological models

Flowering was the phenophase best predicted by the published models. The two GDD models evaluated performed similarly (Figure 1). EF values were positive for both models, i.e. better than the null model, and ranged from 0.2 – 0.53, across the four varieties. An EF > 0.5 is considered to indicate sufficient model quality (Parker et al. 2020b), explaining more than 50% of the variance. Only the GDD 10, 1 VL model for Chardonnay met this criteria.

Model bias (MBE) ranged from -3.4 to 6.5 days with the GDD 0, 60 PK model bias being greater than that for the GDD 10, 1 VL model.

3.1.3. Veraison for published phenological models

The GDD 0, 60 PK models were superior to GDD 10, 1 VL models for veraison, with smaller RMSE for all four varieties (Figure 1). RMSE ranged from 8.5 to 10.9 days for the GDD 0, 60 PK model and 11.3 to 17.6 days for the GDD 10, 0 VL model (Figure 1). MBE ranged from 0.1 to 10.2 days, and were closer to zero for the GDD 0, 60 PK model than the GDD 10, 1 VL model for all varieties except Shiraz. The positive MBE indicated that veraison was predicted later than observed dates for all varieties and models (Figure 1). EF were all positive and were above 0.5 for GDD 0, 60 PK model for all varieties except Cabernet Sauvignon.

3.1.4. Harvest for published phenological models

The harvest models had large errors for Cabernet Sauvignon, Merlot and Shiraz, with RMSE ranging from 17.4 to 27.6 days. For Chardonnay, the RMSE were lower (10.5 – 17.3 days) (Figure 1). EF was only positive for Chardonnay and Shiraz for the two Parker et al. (2020a) models. Otherwise, harvest predictions for other varieties and models were worse than the null model. MBE were mostly large, ≥ 5.8 days, the exception being for Chardonnay predicted by the GDD 0, 91 PK model (Figure 1).

Although there were differences between the structure of the models we evaluated, and differences in how well the model predicted a phenological event, overall, the published models predicted phenology poorly for Australian varieties and environments, particularly for budburst and harvest.

3.2. Update F* of published models

A re-assessment of these models was undertaken using updated F* that had been parameterised using the Australian data (Supplementary Table 2S). At this stage we only updated F*, not temperature thresholds, because we wanted to understand how transferable the published model approach was by just using a localised F*. The data set was split into a parameterisation and validation data set for each variety for this analysis (examples of the data split are shown for Chardonnay and Shiraz in Supplementary Figure 1S). Updated F* were within 12% of the original F* for flowering, veraison and harvest (Supplementary Table 2S). There were much greater differences in F* for budburst. The García de Cortázar-Atauri et al. (2009) models with GDD Tbase 5°C and GDD Tbase 10°C (GDD 5, 1 GCA and GDD 10, 1 GCA) both had large increases in F* of 40% or more. Increases in F* for the van Leeuwen et al. (2008) (GDD 10, 1 VL) model and García de Cortázar-Atauri et al. (2009) (BRIN GCA) model were less than this (Supplementary Table 2S). Note that the results for the updated F* were the same for van Leeuwen et al. (2008) and García de Cortázar-Atauri et al. (2009) models that both used GDD function Tb = 10°C commencing at day 1, given their identical structure (Supplementary Table 2S).

3.2.1. Budburst with updated F* and published models

Although most EFp were positive, EFv were negative for budburst for all models and varieties, other than the BRIN GCA model for Cabernet Sauvignon (EFv = 0.09) (Figure 2). This was therefore the only model in the validation data set that was better than the null model. RMSEv ranged from 7.5 to 17.3 days and almost all RMSE across the parameterisation and validation datasets were lower (better) when updated F* were used compared to the original published model evaluation (compare Figure 1 and 2). MBEv ranged from -2.1 to 2.6 days which indicated a smaller bias when updated F* were used in published models (compare Figure 1 and Figure 2).

3.2.2. Flowering with updated F* and published models

All EFp were positive and ≥ 0.62 (Figure 2) i.e., indicating sufficient model quality (Parker et al. 2020b). However, although EFv were all positive, only the GDD 0, 60 PK model for Chardonnay and Shiraz were > 0.5. RMSE were 8.9 days or less across both datasets, parameterisation and validation, for all models and varieties. RMSEp for the GDD 10, 1 VL models ranged from 0.2 to 1.6 days which is very low and combined with their very high EFp of 0.96 to 1.0, may indicate model overfitting. The suggestion of overfitting for the GDD 10, 1 VL model is supported by the large differences between these values and their corresponding RMSEv and EFv (RMSEv = 6.1 – 8.9 days and EFv = 0.02 – 0.42) (Figure 2). MBEv ranged from -1 to 1.6 days which indicates less bias overall than for the original published models using their F* for flowering models (compare Figure 1 and 2). Overall, using updated F* improved flowering predictions (compare Figure 1 and Figure 2).

3.2.3. Veraison with updated F* and published models

EFp were ≥ 0.72 for both models for all varieties (Figure 2). Only the GDD 0, 60 PK model had EFv ≥ 0.5 for all varieties excluding Cabernet Sauvignon and EFv was negative for the VL GDD 10,1 model for Cabernet Sauvignon. There were similar concerns with overfitting for the GDD 10, 1 VL model with large differences between the RMSE and EF for the parameterisation and validation datasets (Figure 2). For the

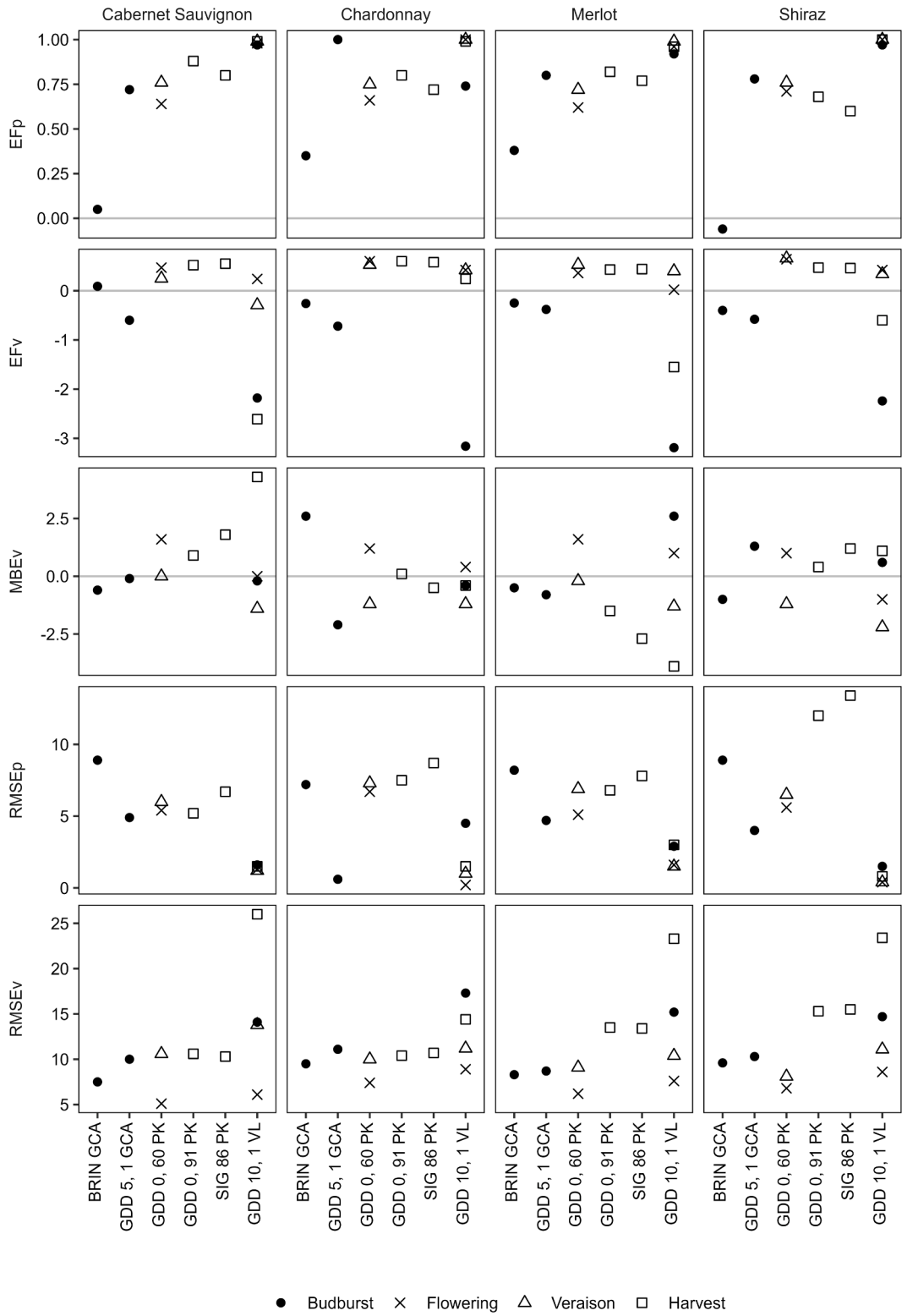


Fig. 2. Performance of published models predicting major phenophase timing using an updated F* with Australian dataset and four varieties, Cabernet Sauvignon, Chardonnay, Merlot and Shiraz. The models were those described by van Leeuwen et al (2008) GDD function Tb = 10°C start day 1 (GDD 10, 1 VL) for all phenophases; Garcia de Cortazar-Atauri et al (2009) GDD function Tb = 5°C, commencing at day 1 (GDD 5, 1 GCA) and Garcia de Cortazar-Atauri et al (2009) BRIN function commencing on day -150 where critical chill (Cc) is satisfied before critical heat (Gc) commences (BRIN GCA) for budburst. Parker et al (2013) GDD Tbase 0°C commencing day 60 model (GDD 0, 60 PK) for flowering and veraison. Parker et al (2020a) GDD Tbase 0°C start day 91 (GDD 0, 91 PK), and Parker et al (2020a) Sigmoid model start day 86 (SIG 86 PK) for harvest. EF; model efficiency, RMSE; root mean square error (days), MBE; mean bias error (days). Subscript p indicates parameterisation dataset and v, the validation dataset. Note that the GDD 10, 1 GCA model is not shown as its results are the same as GDD 10,1 VL model, given their identical structure and therefore the same updated F*.

GDD 0, 60 PK model, RMSEv ranged from 8.1 – 10.6 days. Veraison predictions were overall improved with updated F* (compare Figure 1 and Figure 2) .

3.2.4. Harvest with updated F* and published models

For harvest, EFp were 0.6 or greater. EFv were negative (i.e. worse than the null model) for the GDD 10, 1 VL model for all the red varieties, Cabernet Sauvignon, Merlot and Shiraz. EFv were above or just below 0.5 for all varieties for the GDD 0, 91 PK and SIG 86 PK models. MBEv for all models ranged from -3.9 to 4.3 days, which was much reduced from the original published harvest models. The RMSE was generally large for the updated harvest models ranging up to 26 days, but each updated model RMSEp and RMSEv was lower (better) than the corresponding RMSE for the models that used the published F*, the exception being the RMSEv for Merlot using the GDD 10, 1 VL model. Once again, the GDD 10, 1 VL models had very low RMSEp and high RMSEv suggesting overfitting (Figure 2).

Overall, the updated F* improved the model efficiency and reduced error and bias for the published models. Improvements were not consistent between models, phenophases, nor across varieties. Despite improvements, there were many models with EF below 0.5 i.e. of insufficient quality, concerns with overfitting, some large RMSE, particularly for harvest, and some large differences between the results for the parameterisation and validation data sets, suggesting these models may not translate well to new regions.

3.3. Develop across-variety (species) level models

To further develop models more suited to an Australian context, we trialled new phenological models by first using different heat accumulation methods and fitting all parameters.

There was no single combination of model structure and parameters that was best for all phenological phases (Table 4). The best model for budburst (indicated by lowest AIC) was a Richardson temperature function with a Tbase of 5.6°C and start day of 47; AIC for Sigmoid and GDD functions was similar to that for the Richardson function. EF and RMSE were also similar for these models (Table 4). The BRIN two-phase approach had the highest AIC, EFv was negative and EF were larger than the other budburst models, and it is more complex to use. We did not progress our model development with this two-phase approach.

The Richardson function was the best for flowering, but GDD, Sigmoid and Wang & Engel's functions all had similar AIC, EF and RMSE (Table 4). For veraison and harvest, models using the Sigmoid function were better (lowest AIC). Harvest models performed relatively poorly with a negative value for EF and largest RMSE for the Richardson temperature function. For the other temperature functions, GDD, Sigmoid and Wang and Engel's for harvest, EF was around 0.3 and RMSE, about 17 days for both datasets. Overall, the Sigmoid model was superior across the phenophases when all parameters were fitted with the highest or equal highest EF and the lowest RMSE for all phases except budburst, and MBEv closest to zero for all phenophases (Table 4).

3.4. Develop simplified across-variety (species) level models

To simplify these fully fitted models, we fixed some parameters; the critical temperatures and the start day, when further developing our models (Table 5).

AIC, EF, RMSE and MBE were obtained for a selection of across-variety (species) level models developed that used each of the above temperature response functions. We excluded BRIN and included Single Sine as explained above. Threshold temperature(s) was rounded to the nearest 5°C, in similarity to García de Cortázar-Atauri et al. (2009) and Parker et al. (2020a) and with nominal start days that were within the ideal values obtained from the fully fitted models (Table 4), but that were values that made calculations easier i.e. day 1 (1st July), day 32 (1st August), day 63 (1st September).

The set of models was restricted to show those which could be developed using a single temperature response function with unique threshold temperature(s) but with different start days depending on phenological phase (Table 5).

No model structure and parameter combination performed best for all phenological phases, and generally models that performed well for budburst, did not perform well for the later phenological phases (Table 5). As expected, the AIC was higher when parameters were set (Table 5) than when all parameters were fitted (Table 4), i.e., the models were not as good with some fixed parameters as when all parameters were fitted. For example, the AIC for the best budburst model, Richardson (Tbase 5.6°C, Thigh 44.2°C and start day 47) was 1620 (Table 4), but the AIC for Richardson Tbase 5°C, Thigh 25°C and start day of either 1 or 32 were 1706 or 1632 respectively (Table 5). RMSE for

Table 4
Performance of across-variety (species) level models when all parameters were fitted. AIC; Akaike Information Criterion, EF; model efficiency, RMSE; root mean square error (days), MBE; mean bias error (days). Subscripts denote which dataset - Parameterisation $_P$, or Validation $_V$ was used for the model metric.

Phase	Tbase (°C)	Topt (°C)	Thigh (°C)	StartDay	AIC_P	EF_P	EF_V	$RMSE_P$	$RMSE_V$	MBE_V
GDD function										
Budburst	5.8			47	1621	0.25	0.19	9.7	9.0	-1.0
Flowering	7.8			65	1565	0.67	0.69	6.0	5.9	0.3
Veraison	3.5			51	1916	0.55	0.48	9.3	9.8	-1.0
Harvest	0.0			52	2026	0.30	0.32	17.5	17.2	-1.0
Richardson function										
Budburst	5.6		44.2	47	1620	0.25	0.19	9.7	9.0	-1.0
Flowering	7.8		25.0	65	1562	0.67	0.69	5.9	5.9	0.5
Veraison	3.9		23.5	65	1936	0.52	0.45	9.5	10.0	-3.1
Harvest	5.1		20.0	51	2182	-0.09	-0.09	21.8	21.9	-13.5
Sigmoid function										
Budburst	a = -0.2447, b = 10.41			24	1621	0.25	0.17	9.7	9.2	-0.9
Flowering	a = -0.2638, b = 16.27			51	1563	0.67	0.70	5.9	5.8	0.0
Veraison	a = -0.6128, b = 14.20			78	1856	0.61	0.54	8.7	9.2	-0.8
Harvest	a = -0.4789, b = 13.11			67	2013	0.32	0.35	17.2	16.9	-0.8
Wang and Engel function										
Budburst	-7.6	28.5	40.7	24	1636	0.22	0.15	9.9	9.3	1.8
Flowering	0.4	28.1	37.5	51	1564	0.67	0.69	5.9	5.9	0.7
Veraison	4.0	24.7	34.4	65	1869	0.59	0.54	8.8	9.2	-1.3
Harvest	6.7	27.6	49.9	54	2032	0.30	0.28	17.6	18.5	-5.6
BRIN (chill function then Richardson heat function)										
Budburst	5.0	25.0	-150.0	115	1673	0.13	-0.02	10.5	10.2	0.3

Note: Single Sine was not analysed here as it was not possible to fit a Single Sine model using the analysis package.

Table 5
Performance of some 'single function and threshold temperatures' across-variety (species) level models. Models are grouped by function and threshold temperatures and show options for some phases (e.g. budburst starting on different days). AIC; Akaike Information Criterion, EF; model efficiency, RMSE; root mean square error (days), MBE; mean bias error (days). Subscripts denote which dataset - Parameterisation $_P$, or Validation $_V$ is used for the model metric. Bold indicates the best result for that analysis for that phenophase.

Start Day	Phase	AIC $_P$	EF $_P$	EF $_V$	*RMSE $_P$*	RMSE $_V$	MBE $_V$
Function: GDD, Tbase = 0°C							
1	Budburst * GDD 0, 1	1630	0.23	0.13	**9.8**	9.4	-0.8
32	Budburst * GDD 0, 32	1629	0.23	**0.20**	**9.8**	**9.0**	-0.6
63	Flowering * GDD 0, 63	1703	0.54	0.55	7.0	7.1	0.9
63	Veraison * GDD 0, 63	1933	0.53	0.47	9.5	9.8	-0.3
63	Harvest * GDD 0, 63	2026	0.30	0.32	17.5	17.2	-0.9
1	Flowering	1746	0.50	0.48	7.3	7.6	0.9
1	Veraison	1980	0.47	0.44	10.0	10.1	-0.3
1	Harvest	2027	0.29	0.33	17.5	17.1	**-0.1**
32	Flowering	1697	0.55	0.54	6.9	7.2	1.5
32	Veraison	1946	0.51	0.46	9.6	9.9	-0.8
32	Harvest	2026	0.30	0.32	17.5	17.3	-1.3
Function: GDD, Tbase = 5°C							
1	Budburst	1706	0.05	-0.18	11.0	10.9	-0.3
32	Budburst	1632	0.22	0.15	9.9	9.3	0.5
63	Flowering	1602	0.64	0.64	6.2	6.3	1.1
63	Veraison	1920	0.54	0.45	9.4	10.0	-1.3
63	Harvest	2056	0.23	0.27	18.2	17.8	-1.7
1	Flowering	1741	0.50	0.47	7.3	7.7	1.5
1	Veraison	1989	0.46	0.39	10.1	10.5	-0.9
1	Harvest	2065	0.21	0.27	18.5	17.9	-1.6
32	Flowering	1638	0.61	0.60	6.5	6.7	1.2
32	Veraison	1943	0.52	0.44	9.6	10.1	-1.0
32	Harvest	2062	0.22	0.27	18.4	17.9	-0.7
Function: RICHARDSON, Tbase = 0°C, Thigh = 25°C							
1	Budburst	1630	0.23	0.13	**9.8**	9.4	-0.8
32	Budburst	1629	0.23	**0.20**	**9.8**	**9.0**	-0.6
63	Flowering	1706	0.54	0.55	7.0	7.1	1.0
63	Veraison	1926	0.54	0.49	9.4	9.7	0.2
63	Harvest	2023	0.30	0.32	17.4	17.3	**-0.1**
Function: RICHARDSON, Tbase = 5°C, Thigh = 25°C							
1	Budburst	1706	0.05	-0.18	11.0	10.9	-0.3
32	Budburst *RICH 5-25	1632	0.22	0.15	9.9	9.3	0.5
63	Flowering * RICH 5-25	1602	0.64	0.64	6.2	6.3	1.2
63	Veraison *RICH 5-25	1893	0.57	0.50	9.1	9.6	-0.6
63	Harvest *RICH 5-25	2037	0.27	0.30	17.7	17.5	-1.8
Function: SIGMOID, a=-0.1378, b = 14.76 (best sigmoid for modelling budburst, start day 1)							
1	Budburst * SIG 1	1629	0.23	0.14	**9.8**	9.4	-1.4
Function: SIGMOID, a=-0.1792, b = 16.83 (best sigmoid for modelling budburst, start day 32).							
32	Budburst * SIG 32	**1626**	**0.24**	0.19	**9.8**	**9.0**	-0.9
Function: SIGMOID, a=-0.4274, b = 14.06 (best sigmoid for modelling veraison, start day 63)							
32	Budburst	1704	0.05	-0.05	10.9	10.3	1.3
63	Flowering	1590	**0.65**	**0.67**	**6.1**	**6.1**	0.3
63	Veraison	**1870**	**0.59**	0.52	**8.8**	9.3	-0.4
63	Harvest	2025	0.30	0.33	17.4	17.2	-1.1
Function: SIGMOID, a=-0.4434, b = 13.01 (best sigmoid for modelling harvest, start day 63)							
32	Budburst	1685	0.10	0.01	10.6	10.0	0.5
63	Flowering *SIG 63	1610	0.63	0.64	6.3	6.3	**-0.1**
63	Veraison * SIG 63	1887	0.58	0.52	9.0	9.4	**0.0**
63	Harvest * SIG 63	**2015**	**0.32**	**0.35**	**17.2**	**16.9**	-0.5
Function: WANG and ENGELS, Tbase = 0°C, Topt = 25°C Thigh = 35°C							
1	Budburst	1683	0.11	-0.09	10.6	10.5	0.3
32	Budburst * WE 0-25-35	1635	0.22	0.16	9.9	9.3	-1.2
63	Flowering * WE 0-25-35	1616	0.63	0.65	6.3	6.3	0.2
63	Veraison * WE 0-25-35	1880	0.58	**0.54**	8.9	**9.2**	**0.0**
63	Harvest * WE 0-25-35	2026	0.30	0.31	17.5	17.4	-0.4
Function: WANG and ENGELS, Tbase = 5°C, Topt = 25°C Thigh = 35°C							
1	Budburst	1772	-0.17	-0.51	12.0	12.1	-1.0
32	Budburst	1672	0.10	0.07	10.4	9.7	-0.5
63	Flowering	**1587**	**0.65**	0.66	**6.1**	**6.1**	1.1
63	Veraison	1874	**0.59**	0.53	8.9	9.3	-0.6
63	Harvest	2034	0.28	0.29	17.7	17.6	-1.0
Function: SINGLESINE, Tlower = 5°C, Tupper=25°C							
1	Budburst	1676	0.12	-0.07	10.5	10.4	-0.4
32	Budburst * SS 5-25	1635	0.22	0.16	9.9	9.2	**-0.2**
63	Flowering * SS 5-25	1626	0.62	0.63	6.4	6.5	0.5
63	Veraison * SS 5-25	1899	0.56	0.49	9.1	9.6	-1.0
63	Harvest * SS 5-25	2027	0.29	0.32	17.5	17.2	-0.7
Function: SINGLESINE, Tlower = 5°C, Tupper=30°C							
1	Budburst	1676	0.12	-0.07	10.5	10.4	-0.4
32	Budburst	1635	0.22	0.16	9.9	9.2	**-0.2**
63	Flowering	1607	0.63	0.64	6.2	6.3	0.9
63	Veraison	1902	0.56	0.48	9.2	9.8	-1.0
63	Harvest	2043	0.26	0.29	17.9	17.6	-1.2

Note: To simplify this table, the full set of results for models that were run have not been presented. For example, results for some Sigmoid models for Flowering, Veraison and Harvest were poor so are not shown here. The models we progressed with are marked * and the abbreviations shown correspond with relevant Figure legends.

budburst when some parameters were fixed were 9.0 – 12.1 days (Table 5) compared to 9.0 - 9.9 days for the fully fitted model (Table 4).

Differences in results between the parameterisation and validation datasets were smaller in these across-variety (species) level models when all or some parameters were fitted than when updated F* were used, with RMSEp and RMSEv within 10% for all phenophases when some or all parameters were fitted (Tables 4 and 5). Larger differences than these were observed between RMSEp and RMSEv when updated F* were used (Figure 2).

3.4.1. *Budburst for simplified across-variety (species) level models*

Consideration of each phenophase (Table 5) shows that the best model for budburst was the Sigmoid, a = -0.1792, b = 16.83, with the lowest AIC, the highest EFp (0.24) and second highest EFv. This Sigmoid model also had the equal lowest RMSE of 9.8 and 9.0 days for the parameterization and validation datasets respectively, identical to RMSEp and RMSEv for GDD Tb = 0°C start day 32 and Richardson Tb = 0°C, Thigh = 25°C start day 32 model. MBEv were not the best for this Sigmoid model, but acceptable at -0.9. Three other models, GDD Tb = 0°C, Richardson Tb = 0°C, Thigh = 25°C, and Sigmoid a = -0.1378, b = 14.76 models with a start day of 1 and/or 32 were also strong models for budburst with similar AIC, EF and RMSE to the best Sigmoid model. Single Sine models, start day 32, had the best (closest to zero) MBEv for budburst (Table 5).

3.4.2. *Flowering for simplified across-variety (species) level models*

The best models were the Wang and Engel (WE) Tb = 5°C, Topt = 25°C Thigh = 35°C and Sigmoid a = -0.4274, b = 14.06 models for

flowering, but all flowering models tested were strong, with higher EF overall (all but two above 0.5) and lower RMSE than the other phenophases (Table 5). RMSE for flowering ranged from 6.1 to 7.7 days across these simplified across-variety models.

3.4.3. Veraison for simplified across-variety (species) level models

The Sigmoid a = -0.4274, b = 14.06 model, and both Wang and Engel's models performed strongly for veraison, with the lowest AIC of 1870, 1874 and 1880, respectively. EF ranged from 0.52 – 0.59, RMSE ranged from 8.8 – 9.3 days for these three models and MBEv were -0.4, -0.6 and 0.0 respectively (Table 5).

3.4.4. Harvest for simplified across-variety (species) level models

The Sigmoid a = -0.4434, b = 13.01 was the best model for harvest, but overall, these harvest models were all less accurate than for the other phenophases, with RMSE ranging mostly from 16.9 to 18.5 days and EF around 0.3.

In summary, both the fully parameterised and simplified across-variety models performed better than the published models with the Australian dataset for each phenophase. The simplified models were not quite as strong as the fully parameterised, as expected, but their results were strong enough to counter the complexity concerns of the fully fitted models.

3.5. Develop variety level models

We selected one example from each model type for further analysis. We included a linear GDD model, with an open ended temperature function, to compare it to other models with restricted upper end temperatures, relevant in a climate change setting. GDD Tb = 0°C performed the best of the GDD models trialled across the four phenophases (GDD 0, 1 or 32 for budburst, 63 for other phenophases). The Sigmoid functions performed well depending on their constants and phase, so Sigmoid a = -0.1378, b = 14.76 day 1 (SIG 1), and Sigmoid a = -0.1792, b = 16.83 day 32 (SIG 32) were chosen for budburst, and Sigmoid a = -0.4434, b = 13.01 for modelling other phenophases start day 63 (SIG 63) (Table 5). Although the Sigmoid model chosen was not quite as good overall as that with constants a = -0.4274, b = 14.06 across the three phenophases flowering, veraison and harvest, we prioritised the Sigmoid model that was best for harvest, given that overall, harvest was more poorly predicted than flowering or veraison (Table 5). The Wang and Engel temperature response function places the most limitation on high temperatures, i.e. with Topt = 25°C and Thigh = 35°C, Thigh higher than Richardson or Single Sine functions. The two Wang and Engel's models gave similar results, but that with Tbase = 0°C, Topt = 25°C and Thigh = 35°C (WE 0-25-35) was selected as it was better for budburst and harvest and similar for flowering and veraison (Table 5). The Single Sine temperature response function with a Tlower threshold of 5°C to limit the influence of daily minimum temperature and a Tupper threshold of 25°C to limit the influence of daily maximum temperature (SS 5-25) was not the best model at anything, other than having MBEv closest to zero for budburst, but it performed similarly to the Richardson and Sigmoid function, so was selected. The Single Sine temperature response function produces heat units that accumulate at a reducing rate at higher daily maximum temperature. Richardson Tbase = 5°C and Thigh = 25°C (RICH 5-25) was chosen as it was better for flowering and veraison and similar for harvest and budburst with start day 32 (Table 5).

The performance of these five selected best models where critical temperature and start day was fixed for each of the varieties, Chardonnay, Cabernet Sauvignon, Merlot and Shiraz was evaluated for each phenophase (Figure 3). Sample size (n) and F* used for testing these models are shown in Supplementary Table 4S.

3.5.1. Budburst for variety level models

EFp were mostly 0.5 and above, other than for the GDD Tb = 0°C start day 32 model (GDD 0, 32), for the red varieties, and the Sigmoid model start day 1 (SIG 1) for Cabernet Sauvignon (Figure 3). EFv were all positive but $\leq$ 0.33. Despite the differences in EF between the parameterisation and validation data sets, the RMSE were similar for the two data sets, where RMSEp ranged from 3.8 – 8.0 days and RMSEv ranged from 6.4 to 8.1 days (Figure 3). MBEv ranged from -2.2 to 1.3 days, predicting budburst earlier or later than observed depending on variety and model. Overall, these budburst models were an improvement on previous budburst models with RMSE for both datasets 8.1 days or less (compared to 17.3 days or less for updated F* with published models) and EFp > 0.5 for 23 out of 28 models compared to only 8 out of 24 for previous models. Bias was also slightly reduced (compare Figures 3 and 2).

3.5.2. Flowering for variety level models

All EFp were 0.62 and above (indicating sufficient model quality as mentioned previously), and most EFv were above 0.5, the lowest EFv being 0.35 for Merlot for the GDD 0, 63 model. GDD 0, 63 had the poorest EFv for each variety. RMSEv ranged from 4.7 to 7.5 days. MBEv were all positive, predicting flowering up to 1.8 days later than observed, for all varieties and all models (Figure 3). There were smaller differences between the results for parameterisation and validation data sets for flowering than seen in previous steps of model development. Excluding situations where we had concerns regarding overfitting, these variety level flowering models were an improvement on the published flowering models with updated F* (compare Figure 2 and 3). The GDD 0, 63 was the weakest model for flowering.

3.5.3. Veraison for variety level models

EFp ranged from 0.71 – 0.87 and EFv ranged from 0.24 – 0.70 across the four varieties (Figure 3), with the only EFv < 0.5 being those for Cabernet Sauvignon for all models. RMSEv ranged from 7.5 to 10.6 days and were similar between different models for each variety (Figure 3). All models predicted veraison to be earlier than observed with negative MBEv, with a range of -1.7 to -0.1 days. There was less difference between the results for parameterisation and validation data sets than for previous models that used updated F*. Excluding the GDD 10, 1 VL previous veraison models (due to concerns regarding overfitting), the model statistical metrics were similar or improved for these variety level models when compared to the published models with updated F* (compare Figures 2 and 3).

3.5.4. Harvest for variety level models

EFv were poor (< 0.5) for Merlot and Shiraz for all models, and for Cabernet Sauvignon for the RICH 5-25 and SS 5-25 models. RMSEv ranged from 9.6 to 15.5 days (Figure 3), however, this was an improvement on the RMSEv for the published models with updated F* which ranged from 10.3 to 26 days (Figure 2). Harvest bias ranged from -3.3 to 2.0 days and was greatest for Merlot (Figure 3). All other varieties had a smaller and positive bias with MBEv between zero and 2.0, with Chardonnay having the least bias (Figure 3).

Overall, of these variety-level models, the GDD models generally performed more poorly across the phenophases, with lower EF and higher RMSE than the other models (Figure 3).

A plot of observed versus predicted dates indicates that for earlier observed dates, the predicted results tend to be greater/later than observed i.e. above the 1:1 line, and for later observed dates, the predicted dates tend to be smaller/earlier i.e. below the 1:1 line. That is, there appears to be a systematic bias in these variety-level models (Supplementary Figure 2S), in similarity to instances observed by Parker et al. (2011).

3.6. Develop within season updating models

Further potential improvements to the models were investigated using the concept of within season updating models where actual

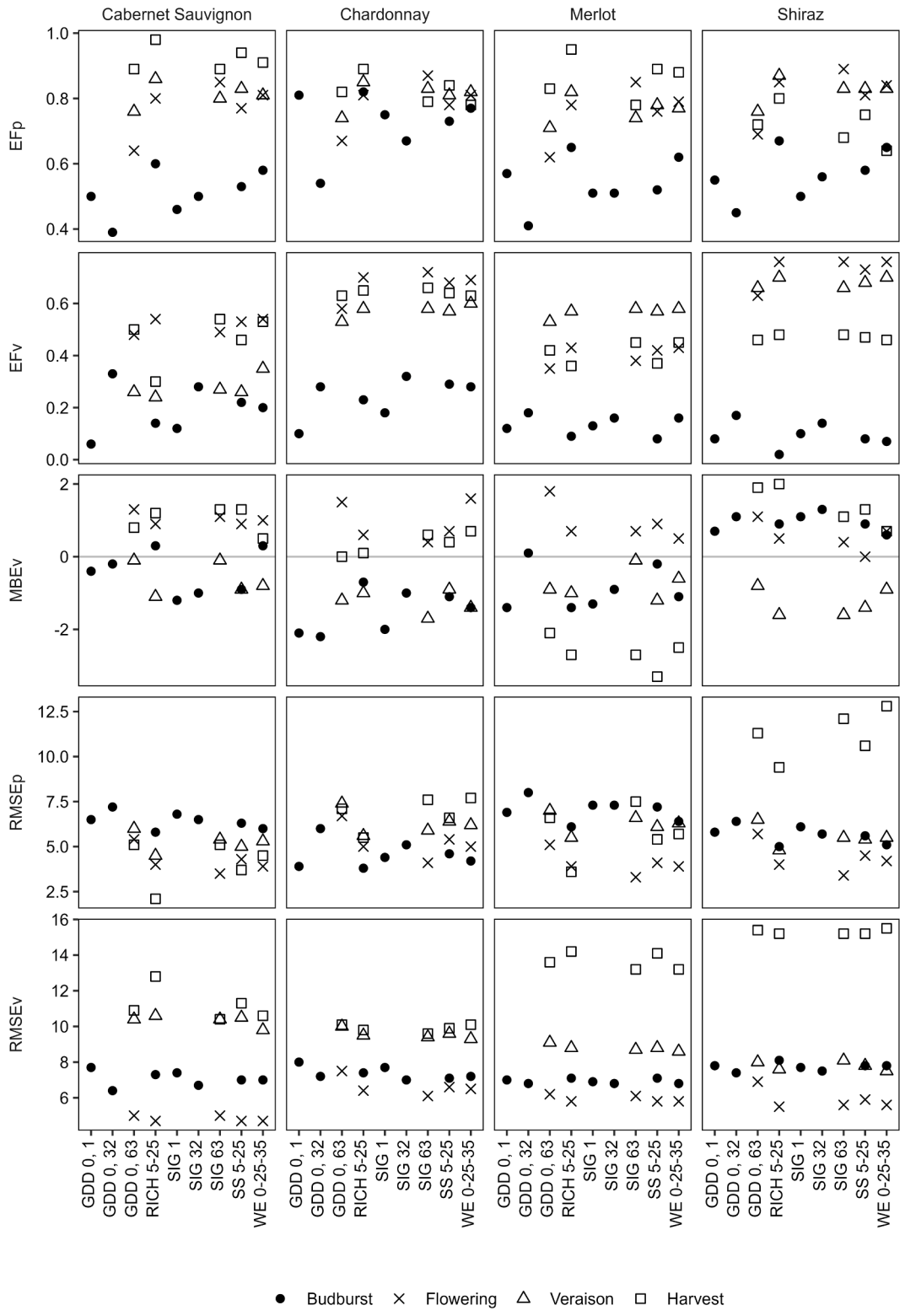


Fig. 3. Performance of variety-level models predicting major phenophase timing with Australian dataset for four varieties, Cabernet Sauvignon, Chardonnay, Merlot and Shiraz. The models evaluated were GDD function Tbase = 0°C commencing day 1 (GDD 0, 1) and GDD function Tbase = 0°C commencing day 32 (GDD 0, 32) for budburst; GDD function Tbase = 0°C commencing day 63 (GDD 0, 63) for flowering, veraison and harvest; Richardson temperature response function Tbase = 5°C and Thigh = 25°C (RICH 5-25); Sigmoid using constants a = - 0.1378, b = 14.76, start day 1 (SIG 1) and a = -0.1792, b = 16.83, start day 32 (SIG 32) for budburst, and a = -0.4434, b = 13.01, start day 63 (SIG 63) for other phenophases; Single Sine Tlower = 5°C and Tupper = 25°C (SS 5-25); Wang and Engel Tbase = 0°C, Topt = 25°C, T high = 35°C (WE 0-25-35) for all phenophases. EF; model efficiency, RMSE; root mean square error (days), MBE; mean bias error (days). Subscript p indicates parameterisation dataset and v, the validation dataset.

phenophase dates within a season were used to predict subsequent phenophases. The data set indicates, for example, that there were 188 instances where we had both a budburst and a flowering date for Shiraz for the same season (Table 1). These data enabled us to predict, for example, veraison from an actual budburst or actual flowering date, or harvest from actual budburst, flowering or veraison dates. The n and F* used for these models are shown in Supplementary Table 6S.

3.6.1. Predicting flowering from actual budburst dates using the variety level models

When flowering was predicted from an actual budburst date, all EFp were 0.79 or greater (Figure 4). EFv were 0.58 or greater for Chardonnay and Shiraz but ranged from 0.16 to 0.35 for Cabernet Sauvignon and were negative for Merlot i.e. poorer than the null model (Figure 4). EFv was superior for Chardonnay when flowering was predicted from an actual budburst date, yet for the red varieties, EFv was superior when flowering was predicted from a set start day (compare Figures 4 and 3).

RMSEp ranged from 1 to 4.5 days for flowering when predicted from an actual budburst date across the four varieties. This compares to a range of 3.3 – 6.7 days when predicted from a set start day (compare Figures 4 and 3). Each model had less error and higher efficiency when predicted from an actual budburst date in the parameterisation data. However, in the validation dataset, it was only for Chardonnay and Shiraz that the error was reduced or similar when flowering was predicted from an actual budburst date.

Model bias ranged from -0.1 to 2.7 days. For Merlot for each model, bias increased when flowering was predicted from an actual budburst date compared to predictions from a set start day (compare Figure 4 and 3). For the other varieties bias was similar or smaller when predicted from an actual budburst date.

In many cases flowering was better predicted from an actual budburst date rather than from a fixed start day. This was however dependent on the variety and the dataset.

3.6.2. Predicting veraison from actual budburst or flowering dates using the variety level models

EFp were all 0.78 and above for veraison predictions from budburst and 0.81 and above when predicted from flowering (Figure 4). Results differed for the validation dataset. EFv were $\geq$ 0.53 for all varieties except Cabernet Sauvignon where EFv ranged from 0.28 to 0.37 when veraison was predicted from flowering, and from -0.02 to 0.07 when veraison was predicted from budburst. In other words, in the validation dataset, model efficiency was sufficient for all varieties except Cabernet Sauvignon, and predictions of veraison from budburst were worse than the null model for the GDD 0, RICH 5-25 and SS 5-25 models for Cabernet Sauvignon. EFv for Chardonnay were similar regardless of the previous phenophase. EFv for Merlot and Shiraz were higher (better) when veraison was predicted from flowering than from budburst (Figure 4).

RMSEv was 5.9 to 11.3 days when veraison was predicted from an actual budburst or flowering date, with less error (lower RMSEv) for all varieties and models, when predicted from flowering than from budburst (Figure 4). A comparison of Figures 3 and 4 indicates that veraison was predicted with less error (lower RMSEv) from flowering than from a set start date. Veraison predictions from budburst generally had more error than predictions from a set start day (compare Figures 3 and 4).

3.6.3. Predicting harvest from actual budburst, flowering or veraison dates using the variety level models

EFp were all $\geq$ 0.68 for all models, varieties and regardless of which phenophase was used to make the prediction from (Figure 4). This result was similar to when harvest was predicted from a set start day (Figure 3). However, when harvest was predicted from a previous phenophase, EFv was particularly poor for Merlot and Cabernet Sauvignon when predicted from veraison with negative EFv for all models (Figure 4). This result is much poorer than when harvest was predicted from a set start day where EFv was positive (0.3 to 0.66) (Figure 3). When harvest was predicted from budburst, EFv ranged from 0.09 to 0.64 and from flowering, from 0.14 to 0.64.

RMSEv ranged from 10.8 to 20.3 days, the poorest values being for Cabernet Sauvignon when predicted from veraison (Figure 4). Model bias MBEv for harvest ranged from -3 to 6.2 days, largest for Merlot (Figure 4). For comparison, model bias ranged from -3.3 to 2 days when harvest was predicted from a set start date (Figure 3).

Our results for within season updating models suggest that models may be improved slightly through adjustments within a vintage year based on knowledge of the current years' observation of the day that the earlier phenological phase was observed, depending on the model used for prediction and the variety. There are however some situations when the RMSE of the validation dataset indicated that large errors may occur in harvest prediction. In most cases, regardless of the previous phenophase, prediction of harvest date remained the most difficult and had the largest RMSE.

Results for minor varieties can be found in Supplementary Tables 3S, 5S and 7S.

4. Discussion

4.1. Evaluate published phenological models

The published models evaluated were mostly one phase GDD linear models with different start days and base temperatures depending on the exact model and phenophase being modelled. One two-phase model was evaluated for budburst, using a chilling then a warming function, and one non-linear sigmoid model was evaluated for harvest (Table 3). We used the F* from the published papers and found that the published models performed poorly with the Australian data (Figure 1). Budburst was particularly poorly predicted with large negative efficiency, indicating that the models were inferior to the null model, meaning that more accurate budburst prediction would be obtained based on mean budburst dates rather than using the predictive model. Errors were large, two weeks or more, and the model bias, particularly for the three García de Cortázar-Atauri et al. (2009) model types, were correspondingly large. The GDD 10, 1 VL model and the GDD 10, 1 GCA, which had the same base temperature and start day, gave quite different outcomes (Figure 1). This was attributed to their different F* and highlighted the importance of F* in model performance and the need for local F* to be derived to improve model performance (Wang et al. 2020). Others have noted that phenology models may need re-calibration when used in different regions or different climates (Drepper et al. 2020, Spafford et al. 2025). We also noted that our results were poorer than the original research. For example, in García de Cortázar-Atauri et al. (2009) original research, which used data from France, the BRIN model for budburst (EL4) had RMSE of 8.2 – 11.6 days across the four varieties Chardonnay, Cabernet Sauvignon, Merlot and Shiraz. Our RMSE ranged from 18.4 – 22.2 days for the same model.

Flowering and veraison predictions with published models were somewhat stronger than for budburst with most efficiencies being positive, i.e., better than the null model. However, with an EF $>$ 0.5 being considered to indicate sufficient model quality (Parker et al. 2020b), only flowering for Chardonnay for the GDD 10, 1 VL model and veraison for Chardonnay, Shiraz and Merlot for the GDD 0, 60 PK model were considered satisfactory by that criteria. In Parker et al. (2013) original research for flowering, model efficiency was superior, between 0.76 and 0.84 for these four varieties. Errors were larger than reported in the original research with Parker et al. (2013) RMSE ranging from 4.8 to 5.6 days, lower than our range of 6.4 to 9.7 days for flowering. For veraison, original Parker et al. (2013) research RMSE ranged from 5.6 to 6.9 days compared to our result of 8.5 to 10.9 days.

Harvest was particularly poorly predicted with negative efficiencies for all varieties for the GDD 10, 1 VL model, and the only positive efficiencies for the Parker GDD and SIG models for Chardonnay and Shiraz

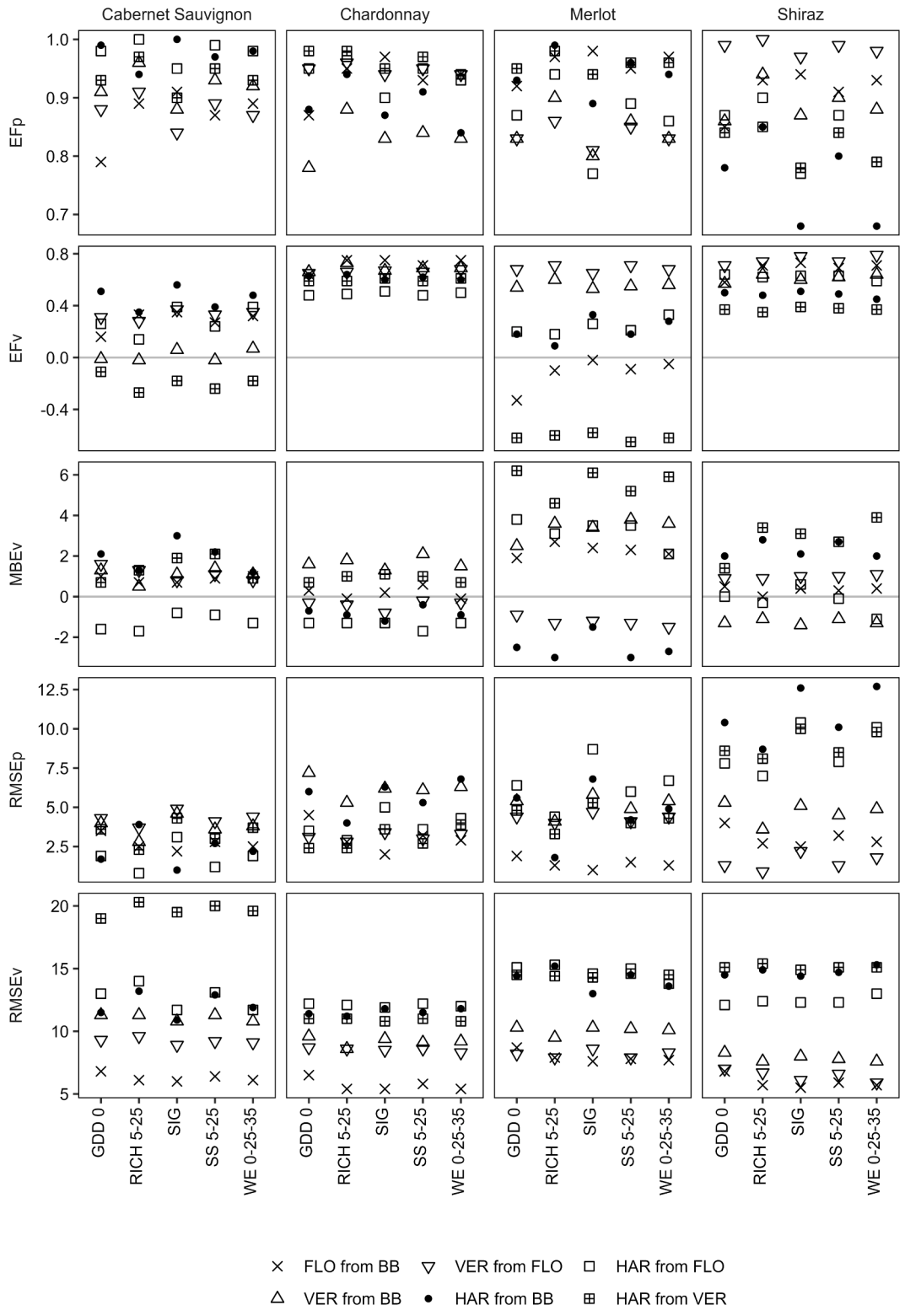


Fig. 4. Performance of within season updating models describing the prediction of later phenophases based on the commencement of F* on the day the earlier phenophase was observed, for four varieties, Cabernet Sauvignon, Chardonnay, Merlot and Shiraz and an Australian dataset. The models evaluated were GDD function Tbase = 0°C (GDD 0) Richardson temperature response function Tbase = 5°C and Thigh = 25°C (RICH 5-25); Sigmoid using constants a = -0.4434, b = 13.01, (SIG); Single Sine Tlower = 5°C and Tupper = 25°C (SS 5-25); Wang and Engel Tbase = 0°C, Topt = 25°C, T high = 35°C (WE 0-25-35). EF; model efficiency, RMSE; root mean square error (days), MBE; mean bias error (days). Subscript p indicates parameterisation dataset and v, the validation dataset. BB; budburst, FLO; flowering, VER; veraison, HAR; harvest.

(Figure 1). Errors were large, ranging from 10.5 to 27.6 days, which were not considered to be adequate, and were larger than those reported in the original research where for example, RMSE ranged from 7 - 10 days in Parker et al. (2020a, 2020b) original models. One explanation for the particularly poor performance of the harvest models with Australian data may be the difficulty in identifying a start day from which to accumulate forcing units. Compared to many grapegrowing regions globally, the Australian viticultural regions are mostly maritime, despite some alpine and inland areas. Therefore, winter is generally relatively mild, meaning that heat units may accumulate in winter, which do not impact phenology due to their early and unseasonal timing. This contrasts with colder winters and relatively rapid transitions to spring in many European continental grapegrowing regions which avoid this early ineffective heat accumulation. Notwithstanding potential explanations for the discrepancies, the published models developed with largely European data, did not perform well with Australian data.

4.2. Update F* of published models

Updating F* using Australian data and re-evaluating the published models produced much improved results. Differences in F* between the published value and that parameterised from Australian data, were particularly large for budburst, and these models were particularly improved with updated F*. The García de Cortázar-Atauri et al. (2009) budburst models had the largest change in F* and the biggest improvement in model performance. Model efficiencies were almost all positive for budburst in the parameterisation dataset and model efficiency (EFp) was superior for the GDD models (EF $>$ 0.5) than for the more complex BRIN model for budburst. Model efficiency was improved but, unsatisfactorily, remained mostly negative in the validation dataset (Figure 2).

While the budburst prediction error (up to 17.3 days) was large, we found that, for example, observed budburst dates for Chardonnay in a given region could vary by up to 46 days (Supplementary Figure 1S), so being able to predict budburst to within 17.3 days can be viewed as an improvement on just making a prediction based on one's knowledge of the possible historic date range for that region and variety.

While improvements in flowering and veraison models using updated F* were not as great as those seen for budburst, the flowering and veraison models were starting from a stronger position. Generally, upon updating F*, the error decreased, model efficiency increased and model bias decreased. The error for flowering prediction was reduced to 8.9 days or less across both the parameterisation and validation datasets. With a possible spread in flowering timing of up to 38 days for a single variety and region (Supplementary Figure 1S), predictions for flowering with errors of under 9 days are clearly useful. Similarly, the largest error for veraison was 13.8 days in the validation data. With a possible spread of veraison dates over time of 48 days in Coonawarra (Supplementary Figure 1S), lower errors from updated veraison models are useful. There were, however, concerns with overfitting for the GDD 10, 1 VL flowering and veraison models.

Harvest predictions also improved with updated F* and published models, with satisfactory EFp, and much reduced model bias. Model error was still high in the validation data set however, and there were concerns for overfitting of the GDD 10, 1 VL harvest model.

In summary, for all phenophases, the models improved overall when updated F* were used; the model efficiency improved, errors decreased, and the models had reduced bias. This was as expected because the models are effectively customised to this Australian data set. Similar findings were made by Wang et al. (2020) for vineyards in some China viticulture regions, where published models were improved when F* were updated using their local data.

Although the models with updated F* were improved, the results obtained were not as good as those reported in the regions where the models were originally developed, with larger errors and with larger differences between results for the parameterisation and validation datasets. Parameterisation results are expected to be better than validation results (Parker et al. 2011) because the data sets are different, despite efforts to make them as similar and balanced as possible (Supplementary Figure 1S) and it is the parameterisation data set that has been used to calibrate the model. Parker et al. (2011) reported that the loss of efficiency between their parameterisation and validation datasets was less than 20% for 7 out of 11 varieties with 4 varieties losing less than 10% efficiency. Our results showed a loss of efficiency of about 10% for flowering for Chardonnay and Shiraz when using the Parker et al. (2013) model, otherwise, our differences in EF values between the parameterisation and validation datasets were mostly 20% or greater. EF $>$ 0.5 is considered to indicate a model with sufficient quality to be useful (Parker et al. 2020b). In our initial assessment of the published models, only 5 out of 44 models tested (4 varieties, 4 budburst, 2 flowering, 2 veraison and 3 harvest for each variety) had EF $>$ 0.5, whereas with the updated F* values, 40 out of 44 EFp and 9 out of 44 models EFv were $>$ 0.5, indicating improvements in the models on this measure. Despite improvements with updated F*, the differences between the EF values for the parameterisation and validation datasets suggest that the model may not perform well more generally, suggesting that further model development was needed.

4.3. Develop across-variety (species) level models (fully fitted and simplified)

Improvements in prediction were obtained by further developing our models. Initially a range of linear and non-linear temperature functions were used with all parameters fitted to identify candidate models for further development. These fully fitted models performed better than those using updated F* and published models for each phenophase (Table 4). Although the linear GDD temperature function was adequate, non-linear temperature functions were in general better, particularly for later phenophases. Non-linear temperature functions have proved superior in aspects of grape phenology (Sadras and Moran 2013) and other crops (Holzworth et al. 2018). Models of budburst resulted in a fitted start day between day 24 and 47 but as one intended use of the winegrape phenology is to examine impacts of a warming climate and possible earlier budburst, it was felt that forcing models for predicting day of budburst commencing on day 32 (i.e. a sensible date, 1st August, that is between the fitted start days obtained) or earlier (preferably day 1) would be more robust. The fitted start day for models of flowering, veraison and harvest obtained were between day 51 (20th August) and day 78 (16th September) (Table 4). Ideal temperatures also varied between the models. The lack of a unique model and parameters that is superior for all phenology phases (Table 3), was not unexpected, nor is it ideal. The model frameworks by van Leeuwen et al. (2008) (GDD Tb = 10°C, start day = 1 used for all phases and varieties), and the Grapevine Flowering Veraison (GFV) model (Parker et al. 2011, 2013) (GDD T*b* = 0°C, start day = 60 for flowering and veraison) have a convenience for users of the model in that the summation of the unique F* for a variety commences at the same day regardless of which of these phenology phases is modelled. It was for these reasons that we then developed models that used the same temperature response functions, and rounded threshold temperature(s) such as 0, 5 or 10°C and with start days corresponding to 1st July (start day 1), 1st August (start day 32) or 1st September (start day 63) (Table 5), rather than those obtained when all parameters were fitted and that were different for each phenophase. As expected, these simplified models were not quite as strong as the fully fitted models, but they performed better than the published models. Therefore, we further developed these simplified models because although the models where all parameters are fitted might theoretically be the best models, they could suffer from over-fitting to the parameterisation set, and their different calculation methods and different values for temperature and start day for each phenophase make them arguably more difficult models to use in a practical sense. In other

words, their accuracy may be outweighed by their complexity and therefore their usefulness.

4.4. Develop variety level models

Though the improvements in these variety-level models were not as dramatic as the difference seen between the original models and those models with updated F*, incremental and meaningful improvements were observed (Figure 3). These improvements were particularly noticeable in the validation dataset results. For example, all efficiencies for all phenophases were positive across both datasets, so all were better than the null model. Errors were more similar between the parameterisation and validation data sets for all phenophases. There were no obvious signs of overfitted models (Figure 3).

For budburst, when updated F* were used, error ranged up to 17.3 days With our models, errors were 8.1 days or less across both the parameterisation and validation data sets, considered to be a useful result. Generally, the GDD 0, 32 model was weaker than the other models, with lower efficiency and higher error in the parameterisation data set. The RICH 5-25 model gave the least error. This pattern however was not replicated for the validation dataset.

For flowering and veraison the GDD 0, 63 model was generally poorer (lower efficiency, higher error) across both data sets, than the other models, which had very similar results (Figure 3). SIG 63 was generally better for flowering and RICH 5-25 better for veraison. These results support those of Wang et al. (2020) who found that non-linear models gave better performance than linear models for flowering and veraison. This suggests that temperature functions with no limits are not the best for these phenophases and that it is necessary to impose an upper temperature limit to improve phenophase modelling. Other than the GDD model, the models were quite tightly grouped for flowering and veraison (Figure 3). The variety-level flowering models compare favourably with the results reported for the original Parker et al. (2013) publication with our error ranging from 3.3 – 6.7 days in the parameterisation data set and 4.7 – 7.5 days in the validation dataset. Parker et al. (2013) error for the same varieties was 4.8 – 5.6 days. The variety-level veraison models had larger errors than the GDD 10, 1 VL models with updated F*, but we had concerns of overfitting for those models. Our veraison results for the variety-level models were approaching those reported in the original Parker et al. (2013) publication with our variety-level models giving errors of 10.6 days or less (7.4 days or less in the parameterisation dataset) where Parker et al. (2013) reported an error of less than one week.

For harvest, the variety-level models gave similar results to the better of the updated F* models and similar results to each other with an error prediction spread of at the most 3.9 days between models (Figure 3). In the parameterisation dataset, the Richardson temperature function was the superior model for harvest overall for each variety and Sigmoid and Wang and Engel 0-25-35 the worst for Chardonnay and Shiraz. Harvest, however, remained a poorly predicted phenophase by these variety-level models.

4.5. Develop within season updating models

Updating prediction with within season actual phenological dates is a common operational and practical use of phenology models in vineyards and this approach gave some further improvements. These were most noticeable for flowering where there were generally improvements in efficiency and error for each model (Figure 4) meaning that it was more accurate to predict flowering from a known budburst date rather than from a fixed start day. Veraison was generally predicted with less error from flowering rather than from budburst (Figure 4). Harvest predictions did not obviously improve with the within season updating models and, in some cases, predictions were poorer, depending on variety (Figure 4), noting that the datasets were not identical. It must also be acknowledged that the data sets used when investigating the models for set start days and within season models are not identical, so it is possible that apparent improvements in predictions from actual phenological dates are a result of the differences between data sets. We did however compare the means, variances and interquartile ranges for the parameterisation and validation data sets used in the different modelling approaches, and they appeared very similar, suggesting further comparisons are unlikely to alter the general findings (results not shown).

Harvest remained the most poorly predicted phenophase with larger errors than the other phenophases, regardless of which phenophase or set start day it was predicted from. This is not surprising. Harvest is pseudo phase (Menzel et al. 2006), it's timing depends on factors such as wine style, regulations and seasonal conditions (Iland et al. 2017), meaning that observed harvest dates may be subject to more variation compared to observed dates for budburst or flowering. Harvest timing could also be influence by crop load (Hannam et al. 2015) which may contribute to the difficulty in predicting this stage of phenology. The poor prediction of harvest presents a problem. Prediction of harvest is arguably the most important, to enable effective vintage logistical planning and to avoid potential problems such as spray residues or overripe fruit (Parker et al. 2020a).

Linear models such as the Richardson, with an upper temperature threshold, or non-linear models, such as the Sigmoid that include the assumption that growth rates vary depending on temperature, are relatively simple to use, and have been shown here to be more robust than open ended linear models. These temperature limiting and non-linear models provide useful predictions of the major phenophases.

There were differences between varieties and their response to phenological prediction. There may be inherent differences between varieties and the interaction between their stage of development or their rate of development during temperature accumulation, or their responsiveness to threshold temperatures (Parker et al. 2011). The different results for different varieties seen in this work may also be influenced by the current data set and exactly which regions the data are sourced from and the weighting of that data between the regions. Although we have tried to capture a range of vineyard regions and climates, we do for example have a larger number of data points for Chardonnay from the relatively cool regions of Adelaide Hills and Coonawarra, and less data from very hot regions such as the Riverland for Shiraz. It is possible that a different dataset would favour different model structures. However, despite this weighting of the data, it is notable that the linear, open-ended GDD function was often poorer than those models with temperature limits or non-linear temperature functions.

Significant improvements were made by developing models for our Australian data set. All models trialled gave improved predictions for the major phenophases, and when season phenophase timing was incorporated into this modelling, further improvements were made. Our work and that of Wang et al. (2020) suggests that local data will not work well with previously published models and that it is important to used locally derived F* (De Rességuier et al. 2026). This suggests there is a need for more data availability for phenophase timing in general, that can be readily drawn on to develop models. Substantial phenological databases exist with European data such as the open access Pan European Phenological database (PEP725) (Templ et al. 2018). In Australia, a lot of phenological data is collected to support management decisions but this is often in private hands and is difficult and time-consuming to access. The development of a national phenological database would be an important step for the further improvement of phenological models suited to Australia.

5. Conclusion

Published phenological models, developed and calibrated for other region-variety combinations, did not perform well with Australian phenological data. A significant improvement was made with re-

parameterised models and updated F* critical. Further improvements were made when a range of heat accumulation methods were used and these improvements were still strong when some parameters, temperature and start day, were fixed to rounded values (which were based on the fitted values), rather than having all parameters fitted. In many cases, phenological prediction was even stronger when in-season predictions were made using previous known phenophase timing. However, these improvements varied depending on the variety, the model and the stage from which the previous phenophase the prediction was based. Overall, harvest was the least accurately predicted with higher error than those for budburst, flowering and veraison. This presents a problem because arguably the accurate prediction of harvest date would be of most use and value to the industry. This research highlighted the need for the collection and documentation of Australian phenophase timing, to assist in the building of future phenological models suited to Australian conditions. Recent and projected warming across all wine regions increases the urgency of data collection and modelling.

CRediT authorship contribution statement

Dane S. Thomas: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Wendy Cameron:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Data curation. **Peter T. Hayman:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Marcos Bonada:** Writing – review & editing, Investigation, Data curation. **Victor O. Sadras:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Joanna M. Gambetta:** Writing – review & editing, Data curation. **Jessica Y.C. Jolley:** Software, Formal analysis. **Paul R. Petrie:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Dr Paul Petrie reports financial support was provided by South Australian Research and Develoment Institute. Paul Petrie reports financial support was provided by Wine Australia. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to acknowledge all the grape growers and researchers who contributed phenology observations for this work. We especially appreciate the work of Dr Richard Hamilton, Mr Chris Brodie, Dr Martin Moran, Mr Gaston Sepulveda and vineyard staff from the Brown Family Wine Group with the provision and collation of this data. This work was jointly funded by the South Australian Research and Development Institute and Wine Australia (project SAR1701-2.2: Managing and modelling vintage compression) whose contribution was derived from levies collected from Australia's grapegrowers and winemakers with matching funds from the Australian Government. The South Australian Research and Development Institute and the University of Adelaide are members of the Wine Innovation Cluster, based at the Waite Campus, Adelaide.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.agrformet.2026.111131.

Data availability

Data will be made available on request.

References

Basler, D., 2016. Evaluating phenological models for the prediction of leaf-out dates in six temperate tree species across central Europe. Agric. For. Meteorol. 217, 10–21. https://doi.org/10.1016/j.agrformet.2015.11.007.

Bernáth, S., Paulen, O., Šiška, B., Kusá, Z., Tóth, F., 2021. Influence of Climate Warming on Grapevine (Vitis vinifera L.) Phenology in Conditions of Central Europe (Slovakia). Plants 10, 1020. https://doi.org/10.3390/plants10051020.

Bock, A., Sparks, T., Estrella, N., Menzel, A., 2011. Changes in the phenology and composition of wine from Franconia, Germany. Clim. Res. 50, 69–81. https://doi.org/10.3354/cr01048.

Bonada, M., Edwards, E.J., McCarthy, M.G., Sepúlveda, G.C., Petrie, P.R., 2020. Impact of low rainfall during dormancy on vine productivity and development. Aust. J. Grape Wine Res. 26, 325–342. https://doi.org/10.1111/ajgw.12445.

Bonada, M., Sadras, V., Thomas, D., Collins, C., Schmidtke, L., Phogat, V., and Petrie, P. 2022, October 26. The relationship between terroir and the phenology of Barossa Shiraz. doi:10.1101/2022.10.25.513684.

Burnham, K.P., Anderson, D.R., 2002. Model selection and multimodel inference: a practical information-heoretic approach. Springer, New York, New York.

Caffarra, A., Eccel, E., 2010. Increasing the robustness of phenological models for Vitis vinifera cv. Chardonnay. Int. J. Biometeorol. 54, 255–267. https://doi.org/10.1007/s00484-009-0277-5.

Caffarra, A., Donnelly, A., Chuine, I., 2011. Modelling the timing of Betula pubescens budburst. II. Integrating complex effects of photoperiod into process-based models. Clim. Res. 46, 159–170. https://doi.org/10.3354/cr00983.

Cameron, W., Petrie, P.R., Barlow, E.W.R., Patrick, C.J., Howell, K., Fuentes, S., 2020. Advancement of grape maturity: comparison between contrasting cultivars and regions. Aust. J. Grape Wine Res. 26, 53–67. https://doi.org/10.1111/ajgw.12414.

Cameron, W., Petrie, P.R., Barlow, E.W.R., 2022. The effect of temperature on grapevine phenological intervals: Sensitivity of budburst to flowering. Agric. For. Meteorol. 315, 108841. https://doi.org/10.1016/j.agrformet.2022.108841.

Chuine, I., Cour, P., Rousseau, D.D., 1999. Selecting models to predict the timing of flowering of temperate trees: implications for tree phenology modelling. Plant Cell Environ. 22, 1–13. https://doi.org/10.1046/j.1365-3040.1999.00395.x.

Chuine, I., García De Cortázar Atauri, I., Kramer, K., Hanninen, H., 2013. Phenology: An integrative environmental science. Plant development models, M.D. Schwartz. Springer, Netherlands, Dordrecht, pp. 275–293.

Chuine, I., 2000. A Unified Model for Budburst of Trees. J. Theor. Biol. 207, 337–347. https://doi.org/10.1006/jtbi.2000.2178.

De Rességuier, L., Parker, A.K., Petitjean, T., De Cortázar-Atauri, I.G., Van Leeuwen, C., 2026. Modeling grapevine phenology at local scale in the context of climate change: An example in the Bordeaux area. Agric. For. Meteorol. 376, 110831. https://doi.org/10.1016/j.agrformet.2025.110831.

Drepper, B., Gobin, A., Remy, S., Van Orshoven, J., 2020. Comparing Apple and Pear Phenology and Model Performance: What Seven Decades of Observations Reveal. Agronomy 10, 73. https://doi.org/10.3390/agronomy10010073.

Duchêne, E., Schneider, C., 2005. Grapevine and climatic changes: a glance at the situation in Alsace. Agron. Sustain. Dev. 25, 93–99. https://doi.org/10.1051/agro:2004057.

Duchêne, E., Huard, F., Dumas, V., Schneider, C., Merdinoglu, D., 2010. The challenge of adapting grapevine varieties to climate change. Clim. Res. 41, 193–204. https://doi.org/10.3354/cr00850.

Eichorn, V.K.W., Lorenz, D.H., 1977. Phänologische Entwicklungsstadien der Rebe. Nachrichtenbl. Deut. Pflanzenschutzd. 29, 119–120.

Essling, M., McKay, S., Petrie, P.R., 2021. Fungicide programs used to manage powdery mildew (Erysiphe necator) in Australian vineyards. Crop Protection 139, 105369. https://doi.org/10.1016/j.cropro.2020.105369.

Fraga, H., Santos, J.A., Moutinho-Pereira, J., Carlos, C., Silvestre, J., Eiras-Dias, J., Mota, T., Malheiro, A.C., 2016. Statistical modelling of grapevine phenology in Portuguese wine regions: observed trends and climate change projections. J. Agric. Sci. 154, 795–811. https://doi.org/10.1017/S0021859615000933.

García De Cortázar-Atauri, I., Brisson, N., Gaudillere, J.P, 2009. Performance of several models for predicting budburst date of grapevine (Vitis vinifera L.). Int. J. Biometeorol. 53, 317–326. https://doi.org/10.1007/s00484-009-0217-4.

García De Cortázar-Atauri, I., Duchêne, E., Destrac-Irvine, A., Barbeau, G., De Rességuier, L., Lacombe, T., Parker, A.K., Saurin, N., Van Leeuwen, C., 2017. Grapevine phenology in France: from past observations to future evolutions in the context of climate change. OENO One 51, 115. https://doi.org/10.20870/oeno-one.2016.0.0.1622.

Godwin, D.C., White, R.J.G., Sommer, K.J., Walker, R.R., Goodwin, I., Clingeleffer, P.R., 2002. VineLOGIC - a model of grapevine growth development and water use. Australian Society of Viticulture and Oenology, Mildura, Victoria, pp. 46–50.

Goodwin, I., 2009. Managing soil and water to target quality and reduce environmental impact. Victorian Department of Primary Industries, Tatura, Victoria, Australia.

Hänninen, H., 1990. Modelling bud dormancy release in trees from cool and temperate regions. Acta Forestalia Fennica 0. https://doi.org/10.14214/aff.7660.

Hannam, K.D., Neilsen, G.H., Neilsen, D., Bowen, P., 2015. Cluster thinning as a tool to hasten ripening of wine grapes in the Okanagan Valley, British Columbia. Can. J. Plant Sci. 95, 103–113. https://doi.org/10.4141/cjps2013-397.

Holzapfel, B., 2009. Management of carbohydrate reserve accumulation as a tool for regulating vine productivity and fruit quality. National Wine Grape Industry Centre, Wagga Wagga, New South Wales, Australia.
Holzworth, D., Huth, N.I., Fainges, J., Brown, H., Zurcher, E., Cichota, R., Verrall, S., Herrmann, N.I., Zheng, B., Snow, V., 2018. APSIM Next Generation: Overcoming challenges in modernising a farming systems model. Environ. Model. Softw. 103, 43–51. https://doi.org/10.1016/j.envsoft.2018.02.002.
Hunter, A.F., Lechowicz, M.J., 1992. Predicting the timing of budburst in temperate trees. J. Appl. Ecol. 29, 597. https://doi.org/10.2307/2404467.
Iland, P., Gago, P., Caillard, A., Dry, P., 2017. Australian Wine - styles and tastes - people and places. Patrick Iland Wine Promotions, South Australia, Australia.
Jeffrey, S.J., Carter, J.O., Moodie, K.B., Beswick, A.R., 2001. Using spatial interpolation to construct a comprehensive archive of Australian climate data. Environ. Model. Softw. 16, 309–330. https://doi.org/10.1016/S1364-8152(01)00008-1.
Kramer, K., 1994. Selecting a model to predict the onset of growth of Fagus sylvatica. J. Appl. Ecol. 31, 172. https://doi.org/10.2307/2404609.
Kriedemann, P.E., 1968. Photosynthesis in vine leaves as a function of light intensity, temperature and leaf age. Vitis 7, 213–220.
Kuhn, M., 2008. Building Predictive Models in R using the Caret package. J. Stat. Softw. 28, 1–26. https://doi.org/10.18637/jss.v028.i05.
Malheiro, A.C., Campos, R., Fraga, H., Eiras-Dias, J., Silvestre, J., Santos, J.A., 2013. Winegrape phenology and temperature relationships in the Lisbon wine region, Portugal. OENO One 47, 287. https://doi.org/10.20870/oeno-one.2013.47.4.1558.
McCarthy, M.G., 1997. The effect of transient water deficit on berry development of cv. Shiraz (Vitis vinifera L.). Aust. J. Grape Wine Res. 3, 2–8. https://doi.org/10.1111/j.1755-0238.1997.tb00128.x.
McCarthy, M., 2019. Assessing clonal variability in Chardonnay and Shira for future climate change. South Australian Research and Development Institute, Urrbrae, South Australia, Australia.
Menzel, A., Sparks, T.H., Estrella, N., Koch, E., Aasa, A., Ahas, R., Alm-Kübler, K., Bissolli, P., Braslavská, O., Briede, A., Chmielewski, F.M., Crepinsek, Z., Curnel, Y., Dahl, Å., Defila, C., Donnelly, A., Filella, Y., Jatczak, K., Måge, F., Mestre, A., Nordli, Ø., Peñuelas, J., Pirinen, P., Remišová, V., Scheifinger, H., Striz, M., Susnik, A., Van Vliet, A.J.H., Wielgolaski, F., Zach, S., Zust, A., 2006. European phenological response to climate change matches the warming pattern. Glob. Chang. Biol. 12, 1969–1976. https://doi.org/10.1111/j.1365-2486.2006.01193.x.
Metropolis, N., Rosenbluth, A.W., Rosenbluth, M.N., Teller, A.H., Teller, E., 1953. Equation of state calculations by fast computing machines. J. Chem. Phys. 21 (6), 1087–1092. https://doi.org/10.1063/1.1699114.
Mo, Y., Zhang, J., Jiang, H., Fu, Y.H., 2023. A comparative study of 17 phenological models to predict the start of the growing season. Front. For. Glob. Change 5, 1032066. https://doi.org/10.3389/ffgc.2022.1032066.
Moran, M., Petrie, P., Sadras, V., 2019. Effects of late pruning and elevated temperature on phenology, yield components, and berry traits in Shiraz. Am. J. Enol. Vitic. 70, 9–18. https://doi.org/10.5344/ajev.2018.18031.
Murray, M.B., Cannell, M.G.R., Smith, R.I., 1989. Date of budburst of fifteen tree species in Britain following climatic warming. J. Appl. Ecol. 26, 693. https://doi.org/10.2307/2404093.
Ortega-Farias, S., Riveros-Burgos, C., 2019. Modeling phenology of four grapevine cultivars (Vitis vinifera L.) in Mediterranean climate conditions. Sci. Horticult. 250, 38–44. https://doi.org/10.1016/j.scienta.2019.02.025.
Parker, A.K., De Cortázar-Atauri, I.G., Van Leeuwen, C., Chuine, I., 2011. General phenological model to characterise the timing of flowering and veraison of Vitis vinifera L.: Grapevine flowering and veraison model. Aust. J. Grape Wine Res. 17, 206–216. https://doi.org/10.1111/j.1755-0238.2011.00140.x.
Parker, A., De Cortázar-Atauri, I.G., Chuine, I., Barbeau, G., Bois, B., Boursiquot, J.M., Cahurel, J.Y., Claverie, M., Dufourcq, T., Gény, L., Guimberteau, G., Hofmann, R.W., Jacquet, O., Lacombe, T., Monamy, C., Ojeda, H., Panigai, L., Payan, J.C., Lovelle, B. R., Rouchaud, E., Schneider, C., Spring, J.L., Storchi, P., Tomasi, D., Trambouze, W., Trought, M., Van Leeuwen, C., 2013. Classification of varieties for their timing of flowering and veraison using a modelling approach: A case study for the grapevine species Vitis vinifera L. Agric. For. Meteorol. 180, 249–264. https://doi.org/10.1016/j.agrformet.2013.06.005.
Parker, A.K., García De Cortázar-Atauri, I., Gény, L., Spring, J.L., Destrac, A., Schultz, H., Molitor, D., Lacombe, T., Graça, A., Monamy, C., Stoll, M., Storchi, P., Trought, M.C. T., Hofmann, R.W., Van Leeuwen, C., 2020a. Temperature-based grapevine sugar ripeness modelling for a wide range of Vitis vinifera L. cultivars. Agric. For. Meteorol. 285–286, 107902. https://doi.org/10.1016/j.agrformet.2020.107902.
Parker, A.K., García De Cortázar-Atauri, I., Trought, M.C.T., Destrac, A., Agnew, R., Sturman, A., Van Leeuwen, C., 2020b. Adaptation to climate change by determining grapevine cultivar differences using temperature-based phenology models: This article is published in cooperation with the XIIIth International Terroir Congress November 17-18 2020, Adelaide, Australia. Guests editors: Cassandra Collins and Roberta De Bei. OENO One 54, 955–974. https://doi.org/10.20870/oeno-one.2020.54.4.3861.
Parmesan, C., 2007. Influences of species, latitudes and methodologies on estimates of phenological response to global warming. Glob. Chang. Biol. 13, 1860–1872. https://doi.org/10.1111/j.1365-2486.2007.01404.x.
Petrie, P.R., Sadras, V.O., 2008. Advancement of grapevine maturity in Australia between 1993 and 2006: putative causes, magnitude of trends and viticultural consequences. Aust. J. Grape Wine Res. 14, 33–45. https://doi.org/10.1111/j.1755-0238.2008.00005.x.
Petrie, P.R., Sadras, V.O, 2016. Quantifying the advancement and compression of vintage. Aust. N.Z. Grapegrow. Winemak. 628, 40–41.
Petrie, P.R., Brooke, S.J., Moran, M.A., Sadras, V.O, 2017. Pruning after budburst to delay and spread grape maturity: Delaying pruning and grapevine phenology. Aust. J. Grape Wine Res. 23, 378–389. https://doi.org/10.1111/ajgw.12303.
R-Core-Team, 2018. A language and environment for statistical computing. R foundation for statistical computing, Vienna, Austria.
Richardson, A.E., Seeley, S.D., Walker, D.R., 1974. A Model for Estimating the Completion of Rest for 'Redhaven' and 'Elberta' Peach Trees1. HortScience 9, 331–332. https://doi.org/10.21273/HORTSCI.9.4.331.
Richardson, E.A., Seeley, S.D., Walker, D.R., Anderson, J.L., Ashcroft, G.L., 1975. Pheno-climatography of Spring peach bud development. HortScience 10, 236–237.
Ruml, M., Korać, N., Vujadinović, M., Vuković, A., Ivanišević, D., 2016. Response of grapevine phenology to recent temperature change and variability in the wine-producing area of Sremski Karlovci, Serbia. J. Agric. Sci. 154, 186–206. https://doi.org/10.1017/S0021859615000453.
Sadras, V.O., Moran, M.A., 2013. Nonlinear effects of elevated temperature on grapevine phenology. Agric. For. Meteorol. 173, 107–115. https://doi.org/10.1016/j.agrformet.2012.10.003.
Sadras, V.O., Reynolds, M.P., De, La, Vega, A.J., Petrie, P.R., Robinson, R., 2009. Phenotypic plasticity of yield and phenology in wheat, sunflower and grapevine. Field. Crops Res. 110, 242–250. https://doi.org/10.1016/j.fcr.2008.09.004.
Sommer, K.J., Islam, M.T., Clingeleffer, P.R., 2000. Light and temperature effects on shoot fruitfulness in Vitis vinifera L. cv. Sultana: Influence of trellis type and grafting. Aust. J. Grape Wine Res. 6, 99–108. https://doi.org/10.1111/j.1755-0238.2000.tb00167.x.
Spafford, L., Lever, J.J., Gessler, A., Zweifel, R., Pietragalla, B., Wegner, J.D., Garnot, V.S. F., Sigg, C., Vitasse, Y., 2025. Stability and transferability of broadly trained phenology models in a changing climate. Agric. For. Meteorol. 372, 110685. https://doi.org/10.1016/j.agrformet.2025.110685.
Templ, B., Koch, E., Bolmgren, K., Ungersböck, M., Paul, A., Scheifinger, H., Rutishauser, T., Busto, M., Chmielewski, F.M., Hájková, L., Hodzić, S., Kaspar, F., Pietragalla, B., Romero-Fresneda, R., Tolvanen, A., Vučetič, V., Zimmermann, K., Zust, A., 2018. Pan European Phenological database (PEP725): a single point of access for European data. Int. J. Biometeorol. 62, 1109–1113. https://doi.org/10.1007/s00484-018-1512-8.
Tomasi, D., Jones, G.V., Giust, M., Lovat, L., Gaiotti, F., 2011. Grapevine Phenology and Climate Change: Relationships and Trends in the Veneto Region of Italy for 1964–2009. Am. J. Enol. Vitic. 62, 329–339. https://doi.org/10.5344/ajev.2011.10108.
Tombesi, S., Scalia, R., Connell, J., Lampinen, B., Dejong, T.M., 2010. Fruit development in almond is influenced by early Spring temperatures in California. J. Horticult. Sci. Biotech. 85, 317–322. https://doi.org/10.1080/14620316.2010.11512674.
Treeby, M.T., 2018. The impact of elevated CO_2 and its interaction with elevated temperature on production and physiology of Shiraz. Department of Economic Development, Jobs, Transport and Resources, Irymple, Victoria, Australia.
Tyerman, S., 2015. Integrating the carbon and water economies of grapevine for optimal management in challenging environments. Adelaide University, Glen Osmond, South Australia, Australia.
van Leeuwen, C., Garnier, C., Agut, C., Baculat, B., Barbeau, G., Besnard, E., Bois, B., Boursiquot, J.M., Chuineİ, I., Dessup, T., Dufourcq, T., García De Cortázar Atauri, I., Marguerit, E., Monamy, C., Koundouras, S., Payan, J.C., Parker, A., Renouf, V., Rodrquez-Lovelle, B., Roby, J.P., Tonietto, J., Trambouze, W., 2008. Heat requirements for grapevine varieties is essential information to adapt plant material in a changing climate. Agroscope Changins-Wädenswil, Nyon, Switzerland.
Van Leeuwen, C., Sgubin, G., Bois, B., Ollat, N., Swingedouw, D., Zito, S., Gambetta, G. A., 2024. Climate change impacts and adaptations of wine production. Nat. Rev. Earth Environ. 5, 258–275. https://doi.org/10.1038/s43017-024-00521-5.
Walker, R., Ashley, R., Cooley, N., Pellegrino, A., Blackmore, D., Clingeleffer, P., Edwards, E., Godwin, D.C., White, R.J.G., Benn, D., 2020. VineLOGIC experimental data sets.v4. CSIRO Data Collection.
Wang, E., Engel, T., 1998. Simulation of phenological development of wheat crops. Agricult. Syst. 58, 1–24.
Wang, X., Li, H., García De Cortázar Atauri, I., 2020. Assessing grapevine phenological models under Chinese climatic conditions. OENO One. https://doi.org/10.20870/oeno-one.2020.54.3.3195.
Wang, T.Y., 1960. A critique of the heat unit approach to plant response studies. Ecology 41, 785–790.
Webb, L.B., Whetton, P.H., Barlow, E.W.R., 2007. Modelled impact of future climate change on the phenology of winegrapes in Australia. Aust. J. Grape Wine Res. 13, 165–175. https://doi.org/10.1111/j.1755-0238.2007.tb00247.x.
Webb, L.B., Whetton, P.H., Barlow, E.W.R., 2011. Observed trends in winegrape maturity in Australia. Glob. Chang. Biol. 17, 2707–2719. https://doi.org/10.1111/j.1365-2486.2011.02434.x.
Wine Australia, 2024. Australian wine sector at a glance. Wine Australia. Available from. https://www.wineaustralia.com/market-insights/australian-wine-sector-at-a-glance [accessed 26 November 2025].
Wine Australia, 2025. National Vintage Report. Wine Australia. Available from file:///C:/Users/wendy/Downloads/MI_VintageReport2025_full_WR..pdf [accessed 30 October 2025].
Zalom, F., Godell, L., Wilson, L., Barnett, W., Bently, W., 1983. Degree days: the calculation and use of heat units in pest management. University of California, Division of Agriculture and Natural Resources.